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Hierarchical Game-Guided Tactical Corridor Planning and Control for Competitive Autonomous Racing

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Keywords: Autonomous racing, Tactical corridor planning, Overtaking decision-making.

Abstract

High-speed autonomous racing requires an ego racecar to decide when and where to overtake, reconstruct a feasible local driving region, and execute the resulting trajectory near the handling limits within a narrow track corridor. These requirements are tightly coupled: a tactically attractive pass can be unsafe if the local corridor is not dynamically feasible, whereas a locally feasible trajectory can be competitively ineffective if it ignores the opponent's response and the upcoming track geometry. This paper proposes a hierarchical game-guided tactical corridor planning and control method for wheel-to-wheel autonomous racing. A Stackelberg–iterative-best-response tactical game is formulated in the Frenet frame under track-boundary, curvature, and tire-friction constraints, and its repeated solution is amortized by a game-guided Truncated Quantile Critics policy. The learned policy does not command steering or throttle directly; instead, it generates behavior decisions, overtaking-side selections, speed caps, and time-varying Frenet corridors that are passed to deterministic downstream solvers. Within each tactical corridor, a sparse second-order cone program computes a curvature-regularized path, a forward–backward speed profiler enforces the GGV handling envelope, and a model predictive contouring controller tracks the reference in closed loop. In the high-fidelity A2RL simulator, the proposed method achieves a 94.7% overtaking success rate over 1000 closed-loop laps with no observed collisions or track-boundary violations, outperforming rule-based, non-interactive MPC, and game-theoretic baselines under the same closed-loop execution stack.

1 Introduction

Autonomous racing has become a demanding benchmark for high-speed decision making, trajectory planning, and at-limit vehicle control. Unlike ordinary lane-level driving, wheel-to-wheel racing compresses tactical interaction, geometric feasibility, and vehicle dynamics into the same short time window: the ego vehicle must infer whether an overtake is strategically viable, reserve enough lateral space before the opportunity disappears, and still respect tire-friction and actuator limits while executing the maneuver. A fixed raceline tracker or a single-vehicle local planner can be fast and stable, but it has no explicit mechanism for deciding when the nominal raceline should be abandoned, which side should be used, or how much of the track should be reserved as a recoverable passing corridor. The central problem is therefore not only to generate a collision-free trajectory, but to construct a tactically meaningful feasible set that remains executable by the planner and controller.

To address the coupled challenges of tactical interaction, constrained trajectory generation, and at-limit execution, autonomous racing research has mainly progressed along three directions: behavior planning, local motion planning, and vehicle control. The behavior-planning layer determines high-level racing intentions such as following, defending, and overtaking; the local-planning layer converts these intentions into feasible trajectories within the limited track space; and the control layer ensures accurate execution near the handling limits of the racecar [1,2,7]. At the behavior-planning layer, the ego racecar needs to determine its high-level tactical intention, including whether to follow, shadow, defend, or overtake, and on which side the overtaking maneuver should be attempted. Classical autonomous driving frameworks usually rely on rule-based finite-state machines, maneuver libraries, or hierarchical decision modules to select discrete driving behaviors, which are subsequently converted into continuous trajectories by the motion planner [3,4]. Similar hierarchical ideas have been adopted in autonomous racing systems, where behavior planners switch among raceline following, obstacle avoidance, and overtaking modes according to opponent position, track availability, and racing regulations [5–7, 9]. These rule-based methods are interpretable and easy to integrate into real vehicles, but their tactical performance strongly depends on manually designed thresholds and track-specific tuning. To explicitly model the strategic coupling between competing vehicles, game-theoretic methods formulate autonomous racing as a non-cooperative dynamic game and compute best-response, Stackelberg, Nash-type, or potential-game strategies [10, 11, 13–16]. Such methods provide a principled way to describe interactive racing behaviors, but exact equilibria of continuous dynamic games are generally difficult to obtain in real time, and approximate solvers may be sensitive to opponent cost assumptions and planning-horizon selection. In parallel, reinforcement learning has shown strong capability in learning complex racing tactics from interaction data [17–21]. However, end-to-end policies usually lack interpretability and do not incorporate game-theoretic interaction structures, making it difficult to reason about competitive feasibility under track and vehicle-performance constraints. Therefore, high-speed competitive racing still requires a behavior-planning method that fuses game-theoretic interaction modeling with learning-based tactical adaptation.

At the local motion-planning layer, the selected behavior or tactical intention must be transformed into a smooth, collision-free, and dynamically feasible trajectory within the constrained track space. In autonomous racing, many methods are built upon offline raceline optimization, where a minimum-curvature or minimum-time reference line is first computed from the track geometry and vehicle dynamic limits [22–26]. The resulting raceline is effective for single-vehicle lap-time optimization, but competitive scenarios require the ego vehicle to deviate from the nominal raceline and actively reconstruct the overtaking region when an opponent blocks the apex or occupies the preferred racing line. To handle dynamic obstacles and alternative passing opportunities, existing local planners employ Frenet-frame polynomial sampling, lattice planning, graph search, trajectory libraries, homotopy-based trajectory generation, and corridor-constrained optimization [3–5, 27, 28]. These methods can explicitly encode track boundaries, obstacle-avoidance constraints, and part of the vehicle dynamic constraints, which makes them suitable for real-time racing implementation. Nevertheless, many local planners still treat the opponent mainly as a moving obstacle and rely on the behavior planner to provide the passing side or maneuver mode [12, 39]. As a result, the generated trajectory may be locally collision-free but tactically conservative, especially when the available overtaking corridor changes rapidly due to opponent motion and track curvature. Sampling- and graph-based planners may require a large number of candidate trajectories to cover different passing possibilities, while nonlinear optimization methods may suffer from local minima and high computational burden [6, 9]. These limitations motivate a structured local-planning scheme in which the behavior layer first constructs a time-varying tactical corridor, and the local planner then generates a curvature-regularized path and dynamically feasible speed profile inside this corridor [17, 18].

At the control layer, the planned reference must be executed accurately while the vehicle operates close to its handling limits. Model predictive control and model predictive contouring control are widely used in autonomous racing because they jointly handle trajectory tracking, progress maximization, and vehicle dynamic constraints [10, 29–33]; learning-based, robust, and tube-MPC variants further improve accuracy and closed-loop robustness under model mismatch and friction variation [27, 28, 34–36]. However, the control performance depends heavily on the feasibility and smoothness of the planned reference [8, 38]: if the path ignores curvature regularity, speed feasibility, or tire-friction limits, the controller is forced to compensate for dynamically inconsistent commands, leading to oscillatory actuation or degraded robustness [26]. Behavior planning, local motion planning, and vehicle control should therefore be coordinated within a unified hierarchical framework.

Taken together, existing studies have advanced behavior planning, local motion planning, and vehicle control, but these modules are still weakly coupled in high-speed competitive racing. Behavior planners often choose a maneuver without exposing an explicit feasible overtaking region to the local planner; local planners can avoid a predicted opponent but usually do not encode overtaking timing or side commitment; and controllers can only track what the planner provides, so curvature discontinuities or speed profiles outside the handling envelope directly degrade closed-loop robustness. This gap motivates a hierarchy in which the tactical layer publishes a time-varying feasible corridor, the planner converts that corridor into a dynamically admissible path–speed reference, and the controller executes the reference while respecting actuator and vehicle-dynamics constraints.

This paper proposes a hierarchical game-guided tactical corridor planning and control method for competitive autonomous racing. The method amortizes a Stackelberg–IBR tactical game with a Truncated Quantile Critics policy, reconstructs the overtaking feasible region as time-varying

Frenet corridors, and combines sparse second-order cone programming, forward–backward speed planning, and model predictive contouring control for dynamically feasible closed-loop execution. The main contributions are:

1. A game-guided reinforcement learning method for tactical corridor synthesis, in which a Stackelberg–IBR tactical game provides a structured interaction model and structural prior, while a Truncated Quantile Critics policy amortizes the equilibrium solution to generate behavior decisions, overtaking-side selections, and time-varying Frenet tactical corridors under track-geometry and vehicle-performance constraints.

2. A curvature-regularized trajectory planning method under tactical-corridor constraints, where a sparse second-order cone program generates feasible paths and a forward–backward speed strategy enforces vehicle performance limits, connecting feasible-region reconstruction with dynamically feasible trajectory generation.

3. A model predictive contouring control method that jointly regulates contouring error, lag error, inputs, and vehicle dynamic constraints to achieve robust closed-loop tracking of the path–speed reference in complex overtaking scenarios.

The remainder of the paper is organized as follows. Section 2 derives the unified vehicle and Frenet models. Section 3 presents the game-guided tactical layer. Section 4 details the SOCP path planner and the FB speed profiler. Section 5 formulates the MPCC controller. Section 6 reports three closed-loop overtake simulations, a comparative evaluation, and an ablation study. Section 7 concludes.

2 Vehicle, Track and GGV Modeling

Figure 1 summarizes the vehicle and track modeling assumptions adopted in the Frenet-coordinate formulation used by all subsequent layers.

Let the continuous-time controller state and input be

$$\boldsymbol{x} = \left[X,\ Y,\ \psi,\ v_x,\ v_y,\ r,\ \delta,\ s_p\right]^\top, \qquad \boldsymbol{u} = \left[a,\ u_\delta\right]^\top, \tag{1}$$

where (X, Y) is the inertial position, ψ is yaw angle, (v_x, v_y) are body-frame longitudinal and lateral velocities, r is yaw rate, δ is front steering angle, and s_p is an auxiliary progress state used by the simulation framework for lap counting. The input a is the commanded longitudinal acceleration and u_δ is the steering-rate command. The rigid-body kinematics are

$$\dot{X} = v_x \cos\psi - v_y \sin\psi, \quad \dot{Y} = v_x \sin\psi + v_y \cos\psi, \quad \dot{\psi} = r, \quad \dot{s}_p = v_x. \tag{2}$$

The front and rear slip angles are

$$\alpha_f = \arctan\left(\frac{v_y + l_f r}{v_x + \varepsilon_v}\right) - \delta, \tag{3}$$

$$\alpha_r = \arctan\left(\frac{v_y - l_r r}{v_x + \varepsilon_v}\right), \tag{4}$$

where the small positive ε_v regularizes the low-speed denominator. With linear lateral tire stiffness,

$$F_{yf} = -C_f \alpha_f, \qquad F_{yr} = -C_r \alpha_r. \tag{5}$$

The resulting Newton–Euler equations are

$$\dot{v}_x = a + r v_y - \frac{1}{m} F_{yf} \sin\delta, \tag{6}$$

$$\dot{v}_y = \frac{1}{m}\left(F_{yf}\cos\delta + F_{yr}\right) - r v_x, \tag{7}$$

$$\dot{r} = \frac{1}{I_z}\left(l_f F_{yf}\cos\delta - l_r F_{yr}\right). \tag{8}$$

The steering angle evolves as $\dot{\delta} = u_\delta$, where u_δ is the steering-rate input in (1). Equations (2)–(8) define

$$\dot{\boldsymbol{x}} = f_c(\boldsymbol{x}, \boldsymbol{u}), \qquad \boldsymbol{x}_{k+1} = f_d(\boldsymbol{x}_k, \boldsymbol{u}_k), \tag{9}$$

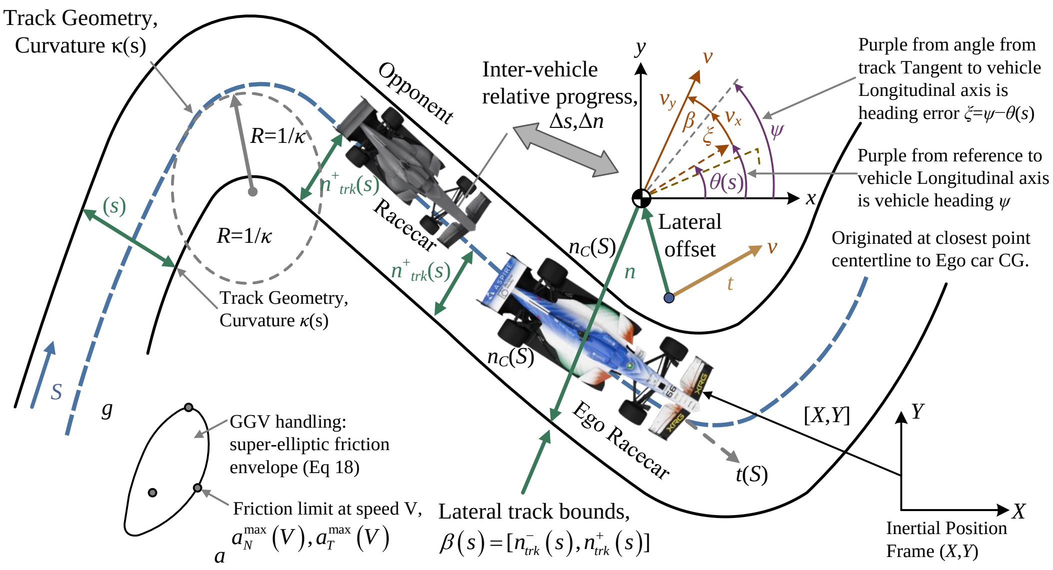


Figure 1: Vehicle and track model in the Frenet frame.

where f_d is the fourth-order Runge–Kutta discretization with sampling period T_s. Numerical parameter values are collected in Table 1.

Let $\boldsymbol{r}_c(s) \in \mathbb{R}^2$ be the track centerline parameterized by arc length $s \in [0, L_{\mathrm{trk}})$. Its tangent angle is $\theta(s)$ and its curvature is $\kappa(s) = d\theta/ds$. Define the tangent and normal unit vectors as

$$\boldsymbol{t}(s) = \big[\cos\theta(s),\ \sin\theta(s)\big]^\top, \qquad \boldsymbol{n}_c(s) = \big[-\sin\theta(s),\ \cos\theta(s)\big]^\top. \tag{10}$$

Any vehicle pose sufficiently close to the track is represented by

$$\boldsymbol{p} = \begin{bmatrix} X \\ Y \end{bmatrix} = \boldsymbol{r}_c(s) + n\,\boldsymbol{n}_c(s), \qquad \xi = \psi - \theta(s), \tag{11}$$

where n is the signed lateral offset and ξ is the heading error. The body velocity projected onto the Frenet axes is

$$v_s = v_x \cos\xi - v_y \sin\xi, \qquad v_n = v_x \sin\xi + v_y \cos\xi. \tag{12}$$

Differentiating (11) gives the exact kinematic relation

$$\dot{s} = \frac{v_s}{1 - \kappa(s)n}, \tag{13}$$

$$\dot{n} = v_n, \tag{14}$$

$$\dot{\xi} = r - \kappa(s)\dot{s}. \tag{15}$$

The wrapped progress difference used in tactical reasoning is always computed on this same coordinate,

$$\Delta s = \big((s_o - s_e + L_{\mathrm{trk}}/2) \bmod L_{\mathrm{trk}}\big) - L_{\mathrm{trk}}/2, \qquad \Delta n = n_o - n_e. \tag{16}$$

For the path planner it is more convenient to use s as the independent variable and to describe a path by $\boldsymbol{z}(s) = [n(s), \xi(s)]^\top$. Let $\omega(s)$ denote the curvature-like steering variable of this geometric path. The spatial Frenet model is

$$\boldsymbol{z}'(s) = \boldsymbol{f}_s\big(\boldsymbol{z}(s), \omega(s); s\big) = \begin{bmatrix} (1-\kappa n)\tan\xi \\ (1-\kappa n)\omega/\cos\xi - \kappa \end{bmatrix}. \tag{17}$$

At planning iteration j, this model is linearized about the previous path iterate $(\bar{\boldsymbol{z}}_k, \bar{\omega}_k)$ at each station s_k,

$$\boldsymbol{z}'(s_k) \approx \boldsymbol{A}_k \boldsymbol{z}_k + \boldsymbol{B}_k \omega_k + \boldsymbol{b}_k, \tag{18}$$

where $\boldsymbol{A}_k$, $\boldsymbol{B}_k$ and $\boldsymbol{b}_k$ are the Jacobian and affine residual of $\boldsymbol{f}_s$. This single linearized Frenet model is the one used by the SOCP in Sec. 4; no new lateral or heading variables are introduced later.

The speed planner and MPCC share the same handling-limit interpretation. For a speed V, the identified GGV table returns the available tangential and normal acceleration limits $a_T^{\max}(V)$ and $a_N^{\max}(V)$. Combined slip is approximated by the super-elliptic envelope

$$\left|\frac{a_N}{a_N^{\max}(V)}\right|^p + \left|\frac{a_T}{a_T^{\max}(V)}\right|^p \leq 1, \tag{19}$$

with exponent $p = 0.75$. Given a path curvature κ_p and speed V, the normal acceleration is $a_N = \kappa_p V^2$, and the remaining tangential acceleration is

$$a_T^{\text{avail}}(V, a_N) = a_T^{\max}(V)\left[1 - \left(\frac{|a_N|}{a_N^{\max}(V)}\right)^p\right]^{1/p}. \tag{20}$$

Aerodynamic drag is modeled as

$$a_{\text{drag}}(V) = \frac{1}{m}\big(c_{d,2}V^2 + c_{d,1}V + c_{d,0}\big), \tag{21}$$

and is added with the acceleration or braking sign convention stated in Sec. 4. The envelope (19) is not a reward term; it is a physical admissibility model used by both the speed profiler and the controller.

3 Tactical Corridor Synthesis

The tactical layer is a game-guided learned decision module with a deterministic optimization interface. Its design is motivated by a Stackelberg–IBR tactical game (Sec. 3.1) whose repeated online solution is amortized by a game-guided TQC policy (Sec. 3.1). The policy outputs a low-dimensional parameter vector encoding the behavior mode, overtaking side, and corridor-shaping parameters, from which a deterministic carver constructs the time-varying Frenet corridor. Let $\mathcal{B}(s) = [n_{\text{trk}}^-(s), n_{\text{trk}}^+(s)]$ denote the track boundary in the Frenet coordinates of Sec. 2, and let $\mathcal{O}_t$ denote the observed opponent set. Online operation is written as the composition

$$\begin{aligned}
\tilde{\boldsymbol{o}}_t &= \mathcal{E}\big(\boldsymbol{x}_{e,t}, \mathcal{O}_t, \mathcal{P}, m_t, \sigma_t^{\text{lock}}\big),\\
\boldsymbol{u}_t &\sim \pi_\theta(\cdot|\tilde{\boldsymbol{o}}_t),\\
\boldsymbol{a}_t &= \Pi_{\mathcal{A}}\left(\underline{\boldsymbol{a}} + \frac{\boldsymbol{u}_t + \mathbf{1}}{2} \odot (\overline{\boldsymbol{a}} - \underline{\boldsymbol{a}})\right),\\
m_{t+1} &= \mathcal{F}_m(\boldsymbol{o}_t, \boldsymbol{a}_t, m_t, \sigma_t^{\text{lock}}),\\
\mathcal{C}_t &= \mathcal{F}_c(\mathcal{B}, \mathcal{O}_t, \boldsymbol{a}_t, m_{t+1}, \sigma_t^{\text{lock}}),\\
\mathcal{G}_t &= (m_{t+1}, V_{\text{cap},t}, \mathcal{C}_t, \pi_t).
\end{aligned} \tag{22}$$

Only π_θ is learned. The projection $\Pi_{\mathcal{A}}$, the finite-state machine $\mathcal{F}_m$, and the corridor carver $\mathcal{F}_c$ are deterministic maps that convert the policy output into auditable constraints for the planner. The training problem is

$$\pi_\theta^\star = \arg\max_{\pi_\theta} \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T_{\text{ep}}-1} \gamma^t r\big(\tilde{\boldsymbol{o}}_t, \boldsymbol{a}_t, m_t, \mathcal{C}_t\big)\right], \tag{23}$$

so reinforcement learning is used to select a tactical feasible set rather than low-level steering or throttle. The proposed algorithmic framework is summarized in Figure 2.

Before presenting the RL implementation, the following establishes the game-theoretic model that motivates the policy structure and supplies its structural prior.

3.1 Game-Guided Tactical Policy

The corridor-synthesis problem is fundamentally interactive: the tactical value of a candidate maneuver depends not only on the currently visible free space, but also on the response it induces in surrounding vehicles and on the terminal planning quality it leaves for subsequent receding-horizon cycles. Therefore, the tactical layer is modeled as a leader–follower (Stackelberg) game in which the ego vehicle (leader) proposes a hybrid tactical action and the surrounding vehicles (followers) respond through a constrained continuous subgame.

Let $\mathcal{N} = \{1, \ldots, N_v\}$ denote the vehicle set and $e \in \mathcal{N}$ the ego. Each vehicle $i \in \mathcal{N}$ selects a hybrid tactical action

$$a_i = (d_i, c_i), \qquad d_i = (m_i, \ell_i), \qquad c_i = (\alpha_i, \rho_i), \tag{24}$$

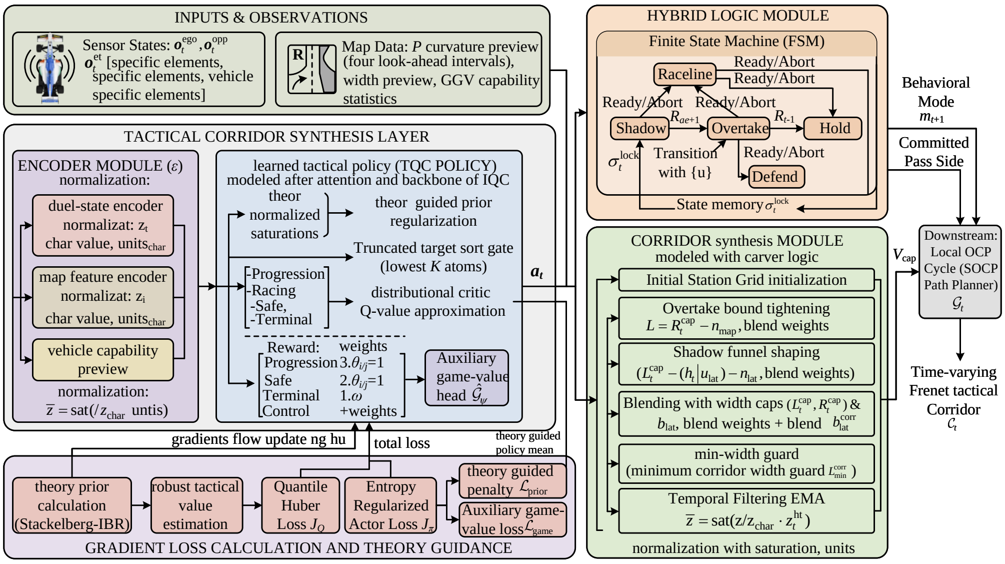


Figure 2: Proposed tactical-stack architecture. The TQC policy amortizes equilibrium-informed tactical decisions while deterministic mappings translate samples into planner-compatible corridor constraints.

where d_i is the discrete component (mode m_i and lateral intention ℓ_i) and c_i is the continuous component (aggressiveness α_i and preference ρ_i). The tactical utility of vehicle i is decomposed as

$$U_i = U_i^{\text{prog}} + U_i^{\text{race}} + U_i^{\text{safe}} + U_i^{\text{term}} + U_i^{\text{ctrl}}, \tag{25}$$

instantiated for the ego as

$$U_e^{\text{prog}} = w_p\, \Delta s_e, \tag{26}$$

$$U_e^{\text{race}} = w_r \sum_{j\in\mathcal{N}_{-e}} (\Delta s_{ej}^{k+H} - \Delta s_{ej}^{k}), \tag{27}$$

$$U_e^{\text{safe}} = -w_s \sum_{j\in\mathcal{N}_{-e}} \phi_s(\Delta_{ej}, \tau_{ej}), \tag{28}$$

$$U_e^{\text{term}} = w_t\, \Phi_{\text{rec}}(x_e^N, \mathcal{R}_e), \tag{29}$$

$$U_e^{\text{ctrl}} = -w_c\, \|a_e^k - a_e^{k-1}\|^2, \tag{30}$$

where Δ_{ej} is an interaction-sensitive separation metric, τ_{ej} a severity indicator, and Φ_{rec} measures the recursive planning quality of the terminal state delivered to the next local OCP cycle. Each follower $i \in \mathcal{N}_{-e}$ maximizes an analogous utility

$$U_i = w_i^p \Phi_i^p + w_i^r \Phi_i^r - w_i^s \Phi_i^s + w_i^t \Phi_i^t - w_i^c \Phi_i^c. \tag{31}$$

Conditioning on a fixed discrete profile $d = (d_e, d_{-e})$, the followers interact over the continuous variables $c_{-e} = \text{col}(c_i)_{i\in\mathcal{N}_{-e}}$ through the constrained response

$$\max_{c_i\in\mathcal{C}_i(d_i)} U_i, \qquad \mathcal{C}_i(d_i) = \{c_i \mid g_i \le 0,\ h_i = 0\}. \tag{32}$$

Define the pseudo-gradient mapping

$$G_c(c_{-e}; d, c_e) = \text{col}\big(-\nabla_{c_i} U_i\big)_{i\in\mathcal{N}_{-e}}. \tag{33}$$

A conditioned continuous follower equilibrium $c_{-e}^{\star}$ satisfies the variational inequality (VI)

$$\big\langle G_c(c_{-e}^{\star}; d, c_e),\ c_{-e} - c_{-e}^{\star} \big\rangle \ge 0, \quad \forall\, c_{-e} \in \mathcal{C}_{-e}(d_{-e}), \tag{34}$$

with $\mathcal{C}_{-e} = \prod_{i\in\mathcal{N}_{-e}} \mathcal{C}_i$. Because the tactical layer must remain synchronous with the receding-horizon planner, an iterative best-response (IBR) approximation replaces the VI solver:

$$c_i^{(\nu+1)} \in \arg\max_{c_i\in\mathcal{C}_i(d_i)} U_i\big(z_i, d, c_e, c_i, c_{-i}^{(\nu)}\big), \tag{35}$$

yielding, at convergence, $c_{-e}^{\mathrm{IBR}\star}(d, c_e) = \lim_{\nu\to\infty} c_{-e}^{(\nu)}$.

To guard against fragile single-response predictions, a local response neighborhood is defined as

$$\mathfrak{R}_{-e}(d, c_e) = \left\{c_{-e} \in \mathcal{C}_{-e} \;\middle|\; \|c_{-e} - c_{-e}^{\mathrm{IBR}\star}\| \le \delta\right\}, \tag{36}$$

and the robust tactical value is

$$\underline{\mathcal{V}}_e(d, c_e) = \min_{c_{-e}\in\mathfrak{R}_{-e}(d,c_e)} U_e\big((d_e, c_e), (d_{-e}, c_{-e})\big). \tag{37}$$

The safe tactical set $\mathcal{A}_e^{\mathrm{safe}}(o_k) \subseteq \mathcal{A}_e$ excludes tactically unsafe or planner-destructive actions. The upper-layer decision is therefore

$$a_e^{k,\star} \in \arg\max_{(d_e,c_e)\in\mathcal{A}_e^{\mathrm{safe}}(o_k)} \underline{\mathcal{V}}_e(d, c_e). \tag{38}$$

Under the standard regularity condition that the follower pseudo-gradient is strongly monotone on a compact convex response set, the follower response is single-valued; with a compact safe action set and continuous tactical value, the upper-level maximizer in (38) is well defined.

Exact online evaluation of (38) is computationally expensive because each candidate action requires follower IBR convergence and terminal quality assessment. The role of the TQC policy π_θ in (22) is therefore to *amortize* repeated solutions of (38) while preserving its strategic semantics. Concretely, the Stackelberg–IBR game supplies three structural priors to the learning problem:

Game-induced reward decomposition. The RL reward mirrors the utility decomposition (25) so that its gradient landscape remains consistent with the upper-layer game.

Game-guided prior policy. The game induces a structured prior over the continuous tactical parameter space. Denote the IBR-consistent tactical parameters by $\boldsymbol{a}^\star_{\mathrm{th}}(\widetilde{\boldsymbol{o}}_t)$. This prior regularizes the learned policy mean through a weighted penalty:

$$\mathcal{L}_{\mathrm{prior}} = \big\|\mu_\theta(\widetilde{\boldsymbol{o}}_t) - \boldsymbol{a}^\star_{\mathrm{th}}(\widetilde{\boldsymbol{o}}_t)\big\|^2_{W_c}, \tag{39}$$

where μ_θ is the deterministic policy mean and W_c is a diagonal weighting matrix that emphasizes safety-critical components (gap thresholds, clearance) over preference parameters (lateral bias).

Auxiliary game-value head. An auxiliary critic head $\hat{\mathcal{G}}_\psi(\widetilde{\boldsymbol{o}}_t, \boldsymbol{a}_t)$ is trained to approximate the robust tactical value:

$$\mathcal{L}_{\mathrm{game}} = \big(\hat{\mathcal{G}}_\psi(\widetilde{\boldsymbol{o}}_t, \boldsymbol{a}_t) - \underline{\mathcal{V}}_e(\boldsymbol{a}_t \mid \widetilde{\boldsymbol{o}}_t)\big)^2. \tag{40}$$

This head provides an auxiliary credit-assignment signal and keeps the learned Q-value anchored to the interaction-aware tactical game, rather than drifting toward an experience-biased value estimate.

Safe masking. In the game formulation, $\mathcal{A}_e^{\mathrm{safe}}(o_k)$ excludes inadmissible tactical actions. In the implementation, this is realized through the projection $\Pi_{\mathcal{A}}$ in (54) and the readiness gate of the FSM: a mode transition to OVERTAKE is blocked unless the corridor-width and GGV-speed admissibility conditions are simultaneously satisfied. Formally, the effective safe policy satisfies

$$\tilde{\pi}_\theta(\boldsymbol{u} \mid \widetilde{\boldsymbol{o}}_t) \propto \pi_\theta(\boldsymbol{u} \mid \widetilde{\boldsymbol{o}}_t)\, \mathbf{1}\big[\Pi_{\mathcal{A}}(\boldsymbol{u}) \in \widehat{\mathcal{A}}_e^{\mathrm{safe}}(\widetilde{\boldsymbol{o}}_t)\big], \tag{41}$$

which filters sampled corridor parameters that would otherwise trigger infeasible or collision-prone FSM transitions.

The overall TQC training objective combines the standard distributional Bellman loss with the game-derived regularizers:

$$\mathcal{L} = J_Q(\varphi) + \lambda_\pi J_\pi(\theta) + \lambda_p \mathcal{L}_{\mathrm{prior}} + \lambda_g \mathcal{L}_{\mathrm{game}}, \tag{42}$$

where J_Q and J_π are the quantile-Huber critic loss and the entropy-regularized actor loss defined below. The learning problem therefore remains structurally anchored to the Stackelberg–IBR tactical game: TQC amortizes the equilibrium computation, the game supplies the structural prior, and safe masking preserves admissibility—so that the resulting corridor policy is simultaneously data-efficient, interaction-aware, and planner-compatible.

3.2 Observation Encoding and Map-Conditioned Features

The remaining subsections instantiate the abstract game quantities into concrete policy variables. The tactical step index k used in the game formulation is hereafter written as t to emphasize real-time operation. The abstract observation o_k is realized as $\widetilde{\boldsymbol{o}}_t$; the discrete component (m_t, σ_t) corresponds to the FSM mode and committed pass side; and the continuous component c_t is implemented by the ten-dimensional parameter vector $\boldsymbol{a}_t$ through the tanh mapping and projection $\Pi_{\mathcal{A}}$ defined below.

The encoder $\mathcal{E}$ combines a duel-state observation with map-conditioned features that provide curvature, width, and vehicle-capability context. The dynamic observation is

$$\boldsymbol{o}_t^{\text{ego}} = \left[\bar{V}_e,\ \bar{\xi}_e,\ \bar{n}_e,\ \bar{a}_{T,e},\ \bar{a}_{N,e},\ \bar{s}_e\right]^\top, \tag{43}$$

$$\boldsymbol{o}_t^{\text{opp}} = \left[\bar{\Delta}s,\ \bar{\Delta}n,\ \Delta\bar{V}_s,\ \bar{s}_o,\ \bar{n}_o,\ \bar{g},\ \bar{\tau}_{\text{tc}},\ \bar{V}_o\right]^\top, \tag{44}$$

$$\boldsymbol{o}_t = \left[\boldsymbol{o}_t^{\text{ego}},\ \boldsymbol{o}_t^{\text{opp}},\ \boldsymbol{b}_t,\ \sigma_t^{\text{lock}},\ \boldsymbol{e}_{m_t}\right]. \tag{45}$$

With five FSM modes, $\boldsymbol{o}_t \in \mathbb{R}^{23}$. Each scalar is normalized as

$$\bar{z} = \text{sat}(z/z_{\text{char}}), \qquad \text{sat}(q) = \min\{1, \max\{-1, q\}\}, \tag{46}$$

using the characteristic values in Table 1. The relative progress and lateral separation are given by (16), $\Delta V_s = V_o - V_e$, and $g = \sqrt{\Delta s^2 + \Delta n^2}$. The clipped time-to-collision feature is

$$\tau_{\text{tc}} = \begin{cases} \text{clip}(\Delta s / \max(-\Delta V_s, \varepsilon_\tau), 0, \tau_{\text{tc,max}}), & \Delta s > 0,\ \Delta V_s < 0, \\ \tau_{\text{tc,max}}, & \text{otherwise}, \end{cases} \tag{47}$$

so only a closing leader creates a finite urgency scale. The bucket vector $\boldsymbol{b}_t$ coarsely encodes whether the ego is fast-closing, speed-matched or receding. The memory variables σ_t^{lock} and $\boldsymbol{e}_{m_t}$ make side commitment and hysteresis observable to the policy.

The dynamic duel state is augmented by a non-trainable map $\mathcal{P} = \{\boldsymbol{\rho}_i\}_{i=0}^{M-1}$. This map contains not only curvature and width, but also a local estimate of vehicle capability. From the GGV envelope in Sec. 2, define the curvature-limited speed preview

$$V_{\text{ggv}}(s) = \sup_{0 \le V \le v_{x,\max}} \left\{V :\ |\kappa(s)|V^2 \le a_N^{\max}(V)\right\}, \tag{48}$$

with $V_{\text{ggv}} = v_{x,\max}$ when the curvature constraint is inactive. For a preview interval $\mathcal{I}(s; d_a, d_b) = [s + d_a, s + d_b]$, the statistics inserted into the policy are

$$\kappa_{d_a:d_b}^{\max}(s) = \max_{\ell \in \mathcal{I}(s;d_a,d_b)} |\kappa(\ell)|, \quad W_{\min}^{L/R}(s) = \min_{\ell \in [s, s+d_p]} |n_{\text{trk}}^{+/-}(\ell)|, \quad V_{\min}^{\text{ggv}}(s) = \min_{\ell \in [s, s+d_p]} V_{\text{ggv}}(\ell). \tag{49}$$

Querying $\mathcal{P}$ at progress s returns a 14-dimensional vector

$$\boldsymbol{\rho}(s) = \big[\underbrace{\bar{\kappa}_0,\ \bar{\kappa}_{0:d_1}^{\max}, \ldots, \bar{\kappa}_{d_3:d_p}^{\max},\ \bar{\kappa}_{\text{mean}},\ \bar{\kappa}_{\text{std}},}_{\text{curvature preview (7)}}\ \underbrace{\bar{W}_{\min}^L,\ \bar{W}_{\min}^R,\ \bar{W}_0^L,\ \bar{W}_0^R,\ \Delta\bar{W}_{\min},}_{\text{width (5)}}\ \underbrace{\bar{V}_0^{\text{ggv}},\ \bar{V}_{\min}^{\text{ggv}}}_{\text{GGV (2)}}\big]^\top. \tag{50}$$

The curvature terms encode upcoming cornering severity at four look-ahead intervals, the width terms encode whether an overtake side remains geometrically open, and the GGV terms encode whether the vehicle can exploit that side without exceeding lateral-acceleration capability. The final actor–critic input is

$$\widetilde{\boldsymbol{o}}_t = \left[\boldsymbol{o}_t, \boldsymbol{\rho}(s_e), \boldsymbol{\rho}(s_o)\right] \in \mathbb{R}^{51}, \tag{51}$$

so an identical relative gap can yield different actions on a straight, at corner entry, or before a narrow exit.

The actor output is not a trajectory but a ten-dimensional tactical parameter vector. A sample $\boldsymbol{u}_t \in [-1, 1]^{10}$ is mapped to

$$\boldsymbol{a}_t = \Pi_{\mathcal{A}}\left(\underline{\boldsymbol{a}} + \frac{1}{2}(\boldsymbol{u}_t + \boldsymbol{1}) \odot (\overline{\boldsymbol{a}} - \underline{\boldsymbol{a}})\right), \tag{52}$$

$$\boldsymbol{a}_t = \left[L_t^{\text{cap}}, R_t^{\text{cap}}, d_{\text{safe}}, V_{\text{cap}}, g_{\text{chase}}, g_{\text{ot}}, g_{\text{abort}}, b_{\text{lat}}, w_{\text{safe}}, q_{\text{side}}\right]^\top. \tag{53}$$

The projection $\Pi_{\mathcal{A}}$ enforces

$$\mathcal{A} = \{\boldsymbol{a} :\ g_{\text{chase}} \ge g_{\text{ot}} + \varepsilon_{co},\ g_{\text{abort}} \ge g_{\text{chase}} + \varepsilon_{ca},\ L_t^{\text{cap}} + R_t^{\text{cap}} \ge \varepsilon_{\text{corr}}\}. \tag{54}$$

The components of $\boldsymbol{a}_t$ have direct downstream meanings: $(L_t^{\mathrm{cap}}, R_t^{\mathrm{cap}})$ cap the left/right corridor width, d_{safe} sets opponent clearance, V_{cap} caps the speed profile, $(g_{\mathrm{chase}}, g_{\mathrm{ot}}, g_{\mathrm{abort}})$ tune FSM hysteresis, b_{lat} sets the far-field lateral bias, w_{safe} controls recovery width, and q_{side} chooses the pass side,

$$\sigma_t = \begin{cases} +1, & q_{\mathrm{side}} \geq 0 \quad \text{left-side pass,} \\ -1, & q_{\mathrm{side}} < 0 \quad \text{right-side pass.} \end{cases} \tag{55}$$

Once σ_t is accepted in Overtake, it is copied into σ_t^{lock} and held until the duel terminates.

The continuous tactical action is converted into a persistent intention by a hysteretic finite-state machine with modes $\mathcal{M} = \{\textsc{Raceline}, \textsc{Shadow}, \textsc{Overtake}, \textsc{Hold}, \textsc{Defend}\}$. Let g_{rear} denote the closest rear gap and $\bar{c}_t$ indicate that the post-overtake immunity window has expired. The selected side is considered ready when the previewed side width and GGV speed are both admissible,

$$\mathrm{ready}_t(\sigma) = \mathbb{1}\left[\min_{s\in[s_e, s_e+d_p]} W_\sigma(s) \geq L_{\min}^{\mathrm{corr}}\right] \mathbb{1}\left[\min_{s\in[s_e, s_e+d_p]} V_{\mathrm{ggv}}(s) \geq \min\{V_e, V_{\mathrm{cap}}\}\right], \tag{56}$$

where W_{+1} and W_{-1} are the left/right widths remaining after opponent exclusion. The mode update is

$$m_{t+1} = \begin{cases} \textsc{Overtake}, & g \leq g_{\mathrm{ot}} \wedge \mathrm{ready}_t(\sigma_t) \wedge \bar{c}_t, \\ \textsc{Shadow}, & g_{\mathrm{ot}} < g \leq g_{\mathrm{chase}}, \\ \textsc{Hold}, & m_t = \textsc{Overtake} \wedge \mathrm{abort}_t, \\ \textsc{Defend}, & g > g_{\mathrm{def}}^{\mathrm{front}} \wedge 0 < g_{\mathrm{rear}} < g_{\mathrm{def}}^{\mathrm{rear}} \wedge \bar{c}_t, \\ \textsc{Raceline}, & g > g_{\mathrm{chase}}, \\ m_t, & \text{otherwise.} \end{cases} \tag{57}$$

The abort flag is

$$\mathrm{abort}_t = \big(g > g_{\mathrm{abort}}\big) \vee \big(\Delta n_{\min} < \Delta n_{\mathrm{safe}} \wedge |\Delta s| < g_{\mathrm{interlock}}\big), \tag{58}$$

where $\Delta n_{\min}$ is the minimum lateral separation predicted within the local interaction window. Therefore the policy adjusts thresholds and side preference, while the hybrid logic preserves a persistent mode sequence for the downstream optimizer.

Figure 3 illustrates the geometric meaning of the FSM modes and the corresponding tactical corridors. The green shaded area denotes the feasible corridor published by the tactical layer, and the green curves denote its lateral boundaries. Raceline and Hold share the same full-track-width corridor; the former encourages lap-time-oriented driving, whereas the latter keeps the corridor open but regulates the speed profile through $V_{\mathrm{cap},t}$ to maintain a safe gap to the preceding vehicle. Shadow, Overtake, and Defend further reshape the corridor according to the opponent position, respectively guiding the ego to the rear-side of the opponent on the chosen passing side in preparation for the pass, supporting side-specific passing, and blocking the opponent's forward driving direction.

The reward is defined on tactical outcome and constraint quality. At each step,

$$\begin{aligned} r_t = {} & w_{\mathrm{ot}}\phi_{\mathrm{ot}} + w_{\mathrm{ovb}}\phi_{\mathrm{ovb}} + w_{\mathrm{col}}\phi_{\mathrm{col}} + w_{\mathrm{lat}}\phi_{\mathrm{lat}} + w_{\mathrm{crush}}\phi_{\mathrm{crush}} + w_{\mathrm{app}}\phi_{\mathrm{app}} \\ & + w_{\mathrm{sm}}\phi_{\mathrm{sm}} + w_{\mathrm{gs}}\phi_{\mathrm{gs}} + w_{\mathrm{sbs}}\phi_{\mathrm{sbs}} + w_{\mathrm{wide}}\phi_{\mathrm{wide}}, \end{aligned} \tag{59}$$

with weights listed in Table 1. The event and geometry features are

$$\phi_{\mathrm{ot}} = \mathbb{1}[\Delta s_{t-1} > 0]\mathbb{1}[\Delta s_t < 0]\mathbb{1}[\bar{c}_t], \tag{60}$$

$$\phi_{\mathrm{ovb}} = \mathbb{1}[\Delta s_{t-1} < 0]\mathbb{1}[\Delta s_t > 0], \tag{61}$$

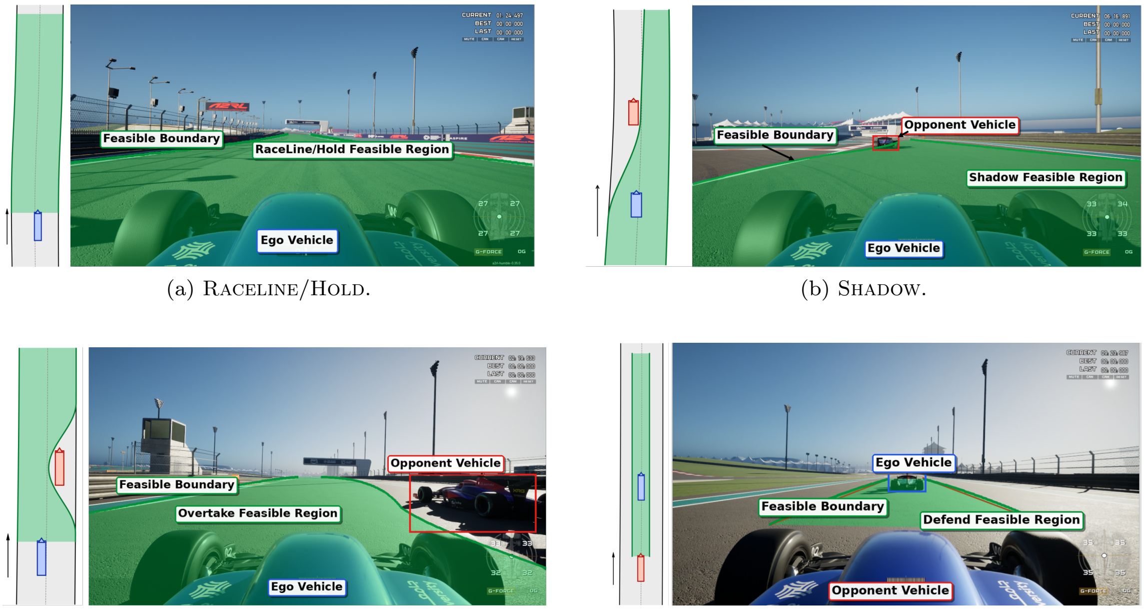
$$\phi_{\mathrm{col}} = \mathbb{1}[\mathrm{contact} \vee \mathrm{offtrack} \vee \mathrm{comm\_loss}], \tag{62}$$

$$\phi_{\mathrm{lat}} = \mathbb{1}[m_t = \textsc{Raceline}]\left(\frac{n_e - n_{\mathrm{rl}}(s_e)}{n_{\mathrm{char}}}\right)^2, \tag{63}$$

$$\phi_{\mathrm{crush}} = \mathbb{1}[|\Delta s| < g_{\mathrm{interlock}}]\left(\frac{\max\{0, \Delta n_{\mathrm{safe}} - |\Delta n|\}}{n_{\mathrm{char}}}\right)^2. \tag{64}$$

The approach feature is

$$\phi_{\mathrm{app}} = \mathbb{1}[\Delta s > 0]\,\mathbb{1}[g_t < g_{\mathrm{chase}}]\,\mathrm{clip}\left(\frac{g_{t-1} - g_t}{g_{\mathrm{char}}}, 0, 1\right), \tag{65}$$



(a) RACELINE/HOLD. (b) SHADOW.

(c) OVERTAKE. (d) DEFEND.

Figure 3: Geometric semantics of mode-dependent tactical corridors. The feasible corridor is released to the full track width in RACELINE/HOLD, funneled toward the chosen passing side behind the opponent in SHADOW, reshaped for side-specific passing in OVERTAKE, and placed in front of the following opponent in DEFEND.

and the regularizers applied to the policy output and the published corridor are

$$\phi_{\text{sm}} = \|\boldsymbol{a}_t - \boldsymbol{a}_{t-1}\|^2_{\boldsymbol{W}_a}, \tag{66}$$

$$\phi_{\text{gs}} = \frac{1}{N_{\text{sta}}} \sum_{k=0}^{N_{\text{sta}}-1} \Big[(n^+_{k,t} - n^+_{k,t-1})^2 + (n^-_{k,t} - n^-_{k,t-1})^2\Big], \tag{67}$$

$$\phi_{\text{sbs}} = \mathbb{1}[|\Delta s| < g_{\text{interlock}}]\mathbb{1}[|\Delta n| \geq \Delta n_{\text{safe}}], \tag{68}$$

$$\phi_{\text{wide}} = \frac{1}{N_{\text{sta}}} \sum_{k=0}^{N_{\text{sta}}-1} \text{clip}\left(\frac{n^+_k - n^-_k - L^{\text{corr}}_{\text{min}}}{n_{\text{char}}}, 0, 1\right). \tag{69}$$

Positive weights reward pass completion, useful gap closing, safe side-by-side tenure and non-degenerate corridors. Negative weights penalize being overtaken, collision, unnecessary raceline deviation, lateral compression, action jitter and corridor jitter. Hence the reward shapes only the tactical interface; it does not modify the SOCP or MPCC objectives. This reward construction preserves the same strategic decomposition that defines the tactical utility (25), ensuring consistency between the game-derived prior and the learning signal.

3.3 Distributional Policy Optimization with TQC

The policy is optimized with Truncated Quantile Critics (TQC), a distributional off-policy algorithm chosen because its pessimistic value estimate reduces the risk of committing to corridors that appear advantageous in expectation but carry high collision probability in the tail. The ablation study in Sec. 6 supports this choice: replacing TQC with standard SAC increases collisions from zero to five. Two critics parameterized by φ_i approximate the return distribution with N_q quantile atoms $\{z^j_{\varphi_i}(\widetilde{\boldsymbol{o}}, \boldsymbol{u})\}_{j=1}^{N_q}$. Given a replay transition, draw $\boldsymbol{u}' \sim \pi_\theta(\cdot|\widetilde{\boldsymbol{o}}')$ and form the truncated target

$$\mathcal{Z}_{\text{targ}} = \text{sort}\Big(\{r_t + \gamma(1 - \mathbb{1}_{\text{done}})[z^j_{\varphi_i}(\widetilde{\boldsymbol{o}}', \boldsymbol{u}') - \alpha \log \pi_\theta(\boldsymbol{u}'|\widetilde{\boldsymbol{o}}')]\}_{i,j}\Big)_{1:K}. \tag{70}$$

Retaining only the lowest K out of $2N_q$ atoms suppresses overestimation of rare high-reward overtakes that coincide with near-miss geometries. The critic minimizes the quantile Huber loss

$$J_Q(\varphi) = \sum_{i=1}^{2} \sum_{j=1}^{N_q} \mathbb{E}_{\mathcal{D}}\left[\rho^{\kappa_q}_{\hat{\tau}_j}\left(y - z^j_{\varphi_i}(\widetilde{\boldsymbol{o}}_t, \boldsymbol{u}_t)\right)\right], \tag{71}$$

where $y \in \mathcal{Z}_{\text{targ}}$, and the actor maximizes the truncated Q-value with entropy regularization:

$$J_\pi(\theta) = \mathbb{E}_{\widetilde{\boldsymbol{o}}\sim\mathcal{D},\boldsymbol{u}\sim\pi_\theta}\left[\alpha\log\pi_\theta(\boldsymbol{u}|\widetilde{\boldsymbol{o}}) - \frac{1}{K}\sum_{z\in\mathcal{Z}_K(\widetilde{\boldsymbol{o}},\boldsymbol{u})} z\right]. \tag{72}$$

The entropy temperature α is tuned automatically. Together with the game-derived regularizers (39)–(40), the combined objective (42) ensures that the learned policy remains a structurally faithful approximation of the Stackelberg–IBR equilibrium. Training episodes randomize initial gap, relative speed, lateral offset and opponent behavior; termination occurs at pass completion, being overtaken, collision, off-track excursion, communication loss or timeout. Evaluation uses the deterministic mean action with the same FSM and corridor carver.

The tactical output consumed by the planner is

$$\mathcal{G}_t = \big(m_{t+1}, V_{\text{cap},t}, \mathcal{C}_t, \pi_t\big), \qquad \mathcal{C}_t = \{(s_k, n_k^-, n_k^+)\}_{k=0}^{N_{\text{sta}}-1}, \tag{73}$$

where π_t stores race-control permission. The carver is the deterministic map

$$\mathcal{C}_t = \mathcal{F}_c\left(\{n_{\text{trk}}^-(s_k), n_{\text{trk}}^+(s_k)\}_{k=0}^{N_{\text{sta}}-1}, \mathcal{O}_t, m_{t+1}, \sigma_t^{\text{lock}}, \boldsymbol{a}_t\right). \tag{74}$$

It first initializes the station grid and physical bounds,

$$s_k = s_e + k\Delta s_{\text{sta}}, \qquad n_k^- = n_{\text{trk}}^-(s_k), \qquad n_k^+ = n_{\text{trk}}^+(s_k). \tag{75}$$

For an opponent with heading error ξ_o and footprint (L_v, W_v), the half-extents projected onto Frenet axes are

$$\rho_n = \frac{1}{2}(L_v|\sin\xi_o| + W_v|\cos\xi_o|), \qquad \rho_s = \frac{1}{2}(L_v|\cos\xi_o| + W_v|\sin\xi_o|), \tag{76}$$

and the lateral exclusion radius is

$$\rho_{\text{exc}} = \rho_n + \frac{W_v}{2} + d_{\text{safe}}. \tag{77}$$

The longitudinal influence of the footprint is

$$\phi_{\text{fade}}(\Delta s_k) = \begin{cases} \cos^{p_{\text{fade}}}\left(\frac{\pi}{2}\frac{|\Delta s_k|}{S_{\text{fade}}}\right), & |\Delta s_k| < S_{\text{fade}}, \\ 0, & \text{otherwise}, \end{cases} \tag{78}$$

multiplied by a startup ramp $\phi_{\text{sr}}(k)$ and a distance ramp $\phi_{\text{rmp}}(k)$. In Overtake, the bound on the opponent side is tightened and the selected side is left open:

$$\begin{cases} n_k^- \leftarrow \max\big(n_k^-,\, n_{o,k} + \rho_{\text{exc}}\phi_{\text{fade}}\phi_{\text{sr}}\phi_{\text{rmp}}\big)\,, & \sigma_t = +1, \\ n_k^+ \leftarrow \min\big(n_k^+,\, n_{o,k} - \rho_{\text{exc}}\phi_{\text{fade}}\phi_{\text{sr}}\phi_{\text{rmp}}\big)\,, & \sigma_t = -1. \end{cases} \tag{79}$$

In Shadow, the interval is shaped into an asymmetric funnel toward σ_t; in Hold, it is relaxed toward the raceline; in Defend, it is biased toward the defensive side. The learned bias and width caps are then blended into the far field. With

$$w_k = \begin{cases} (k/k_{\text{blend}})^2, & k < k_{\text{blend}}, \\ 1, & k \geq k_{\text{blend}}, \end{cases} \tag{80}$$

$$n_k^+ \leftarrow (1-w_k)n_k^+ + w_k\min\big(n_k^+, b_{\text{lat}} + L_t^{\text{cap}}\big), \tag{81}$$

$$n_k^- \leftarrow (1-w_k)n_k^- + w_k\max\big(n_k^-, b_{\text{lat}} - R_t^{\text{cap}}\big). \tag{82}$$

The minimum-width guard is

$$n_k^+ - n_k^- \geq L_{\min}^{\text{corr}}. \tag{83}$$

If this inequality cannot be recovered within the physical track boundary, the previous feasible interval is held at that station. Finally, the published corridor is temporally filtered,

$$\begin{bmatrix} n_k^- \\ n_k^+ \end{bmatrix}_t \leftarrow \beta\begin{bmatrix} n_k^- \\ n_k^+ \end{bmatrix}_t + (1-\beta)\begin{bmatrix} n_k^- \\ n_k^+ \end{bmatrix}_{t-1}, \tag{84}$$

and the tuple $\mathcal{G}_t$ is sent to the SOCP planner. Thus the learned layer contributes exactly three planning quantities: m_{t+1} selects the mode-dependent terminal target, $V_{\text{cap},t}$ clips the GGV speed profile, and $\mathcal{C}_t$ supplies the lateral inequalities $n_k^- \leq n_k \leq n_k^+$ in Sec. 4. If a tactical update is delayed by the simulation interface, the last valid $\mathcal{G}_t$ is held for one planning cycle.

4 Local Planning

The local planner converts the tactical tuple $\mathcal{G}_t = (m_{t+1}, V_{\mathrm{cap},t}, \mathcal{C}_t, \pi_t)$ into a time-parameterized reference. Its structure is deliberately split:

$$\mathcal{G}_t \xrightarrow{\text{ECOS SOCP}} \{s_k, n_k^\star, \xi_k^\star, \kappa_{p,k}^\star\}_{k=0}^{N-1} \xrightarrow{\text{FB speed surface}} \mathcal{R}_t. \tag{85}$$

The first stage solves a sparse second-order cone program for the Frenet path. The second stage computes a forward–backward (FB) speed surface on the resulting local path using the GGV envelope of Sec. 2.

The horizon is sampled at

$$s_k = s_e + k\Delta s_{\mathrm{sta}}, \qquad k = 0, \dots, N-1, \tag{86}$$

where $N = N_{\mathrm{sta}}$ is the number of corridor stations and the wrapped value of s_k is used when querying track data. The tactical corridor is intersected with the physical track boundary,

$$\underline{n}_k = \max\{n_{\mathrm{trk}}^-(s_k), n_k^-\}, \qquad \overline{n}_k = \min\{n_{\mathrm{trk}}^+(s_k), n_k^+\}. \tag{87}$$

If no tactical corridor is active, $\underline{n}_k$ and $\overline{n}_k$ are the track limits after the standard safety margin. The optimization variables are

$$\boldsymbol{y} = \left[\{n_k, \xi_k, \omega_k, \sigma_k^\omega\}_{k=0}^{N-1}, \epsilon_n, \epsilon_\xi\right], \qquad \boldsymbol{z}_k = \begin{bmatrix} n_k \\ \xi_k \end{bmatrix}, \tag{88}$$

where (n_k, ξ_k) are the Frenet variables of Sec. 2, ω_k is the curvature-like path input in (17), σ_k^ω is the cone epigraph for ω_k^2, and $(\epsilon_n, \epsilon_\xi)$ are terminal relaxation variables.

The initial condition is hard,

$$\boldsymbol{z}_0 = \begin{bmatrix} n_e \\ \xi_e \end{bmatrix}, \tag{89}$$

while the terminal target is mode dependent,

$$n_T = \begin{cases} n_{\mathrm{rl}}(s_{N-1}), & m_t = \textsc{Raceline},\ n_{\mathrm{rl}}(s_{N-1}) \in [\underline{n}_{N-1}, \overline{n}_{N-1}], \\ \frac{1}{2}(\underline{n}_{N-1} + \overline{n}_{N-1}), & \text{otherwise}, \end{cases} \qquad \xi_T = 0. \tag{90}$$

Instead of forcing this terminal condition as an equality, the implemented planner uses soft terminal inequalities,

$$\begin{aligned} -\epsilon_n \le n_{N-1} - n_T \le \epsilon_n, \qquad -\epsilon_\xi \le \xi_{N-1} - \xi_T \le \epsilon_\xi, \\ \epsilon_n \ge 0, \qquad \epsilon_\xi \ge 0. \end{aligned} \tag{91}$$

This prevents ECOS infeasibility when a short horizon begins inside a tight overtake corridor but still biases the path toward the tactical terminal intent.

At sequential-linearization iteration j, the nonlinear spatial model (17) is linearized at $(\boldsymbol{z}_k^j, \omega_k^j)$,

$$\boldsymbol{f}_s(\boldsymbol{z}_k, \omega_k; s_k) \approx \boldsymbol{A}_k^j \boldsymbol{z}_k + \boldsymbol{B}_k^j \omega_k + \boldsymbol{b}_k^j \triangleq \hat{\boldsymbol{f}}_{s,k}^j(\boldsymbol{z}_k, \omega_k). \tag{92}$$

Trapezoidal integration gives the linear equality constraint

$$\boldsymbol{z}_{k+1} - \boldsymbol{z}_k = \frac{\Delta s_{\mathrm{sta}}}{2}\left[\hat{\boldsymbol{f}}_{s,k}^j(\boldsymbol{z}_k, \omega_k) + \hat{\boldsymbol{f}}_{s,k+1}^j(\boldsymbol{z}_{k+1}, \omega_{k+1})\right], \quad k = 0, \dots, N-2. \tag{93}$$

The lateral corridor and curvature epigraph are

$$\underline{n}_k \le n_k \le \overline{n}_k, \qquad k = 0, \dots, N-1, \tag{94}$$

$$\left\| \begin{bmatrix} 2\omega_k \\ \sigma_k^\omega - 1 \end{bmatrix} \right\|_2 \le \sigma_k^\omega + 1, \qquad k = 0, \dots, N-1, \tag{95}$$

which is equivalent to $(\omega_k)^2 \le \sigma_k^\omega$ with $\sigma_k^\omega \ge 0$. The SOCP solved by ECOS is

$$\begin{aligned} \boldsymbol{y}^{j+1} = \arg\min_{\boldsymbol{y}} \sum_{k=0}^{N-1} w_\omega \sigma_k^\omega + w_{n,T}\epsilon_n + w_{\xi,T}\epsilon_\xi \\ \text{s.t. (89), (91), (93), (94), and (95).} \end{aligned} \tag{96}$$

Equivalently, ECOS receives the conic form

$$\min_{\boldsymbol{y}} \boldsymbol{c}^\top \boldsymbol{y}, \qquad \boldsymbol{A}_{\rm eq}\boldsymbol{y} = \boldsymbol{b}_{\rm eq}, \quad \boldsymbol{G}_{\rm lp}\boldsymbol{y} \le \boldsymbol{h}_{\rm lp}, \quad \|\boldsymbol{G}_{q,k}\boldsymbol{y} + \boldsymbol{g}_{q,k}\|_2 \le \boldsymbol{h}_{q,k}^\top \boldsymbol{y} + h_{q,k}. \tag{97}$$

After each solve, the linearization trajectory is updated. The loop stops when

$$\Delta_n^j = \max_k |n_k^{j+1} - n_k^j| \le \varepsilon_n^{\rm sl}, \qquad \Delta_\xi^j = \max_k |\xi_k^{j+1} - \xi_k^j| \le \varepsilon_\xi^{\rm sl}, \tag{98}$$

or when the iteration limit is reached. The optimized Cartesian path and its heading are

$$\boldsymbol{p}_k^\star = \boldsymbol{r}_c(s_k) + n_k^\star \boldsymbol{n}_c(s_k), \qquad \psi_k^\star = \theta(s_k) + \xi_k^\star, \qquad \kappa_{p,k}^\star \approx \omega_k^\star. \tag{99}$$

The path arclength used by the speed planner is then recomputed from the optimized Cartesian points,

$$\ell_0 = 0, \qquad \ell_{k+1} = \ell_k + \|\boldsymbol{p}_{k+1}^\star - \boldsymbol{p}_k^\star\|_2, \qquad \Delta\ell_k = \ell_{k+1} - \ell_k. \tag{100}$$

The FB speed profiler computes the largest speed surface that remains consistent with curvature, longitudinal acceleration and braking. First, a curvature speed ceiling is obtained at each station:

$$V_{{\rm lim},k} = \begin{cases} v_{x,\max}, & |\kappa_{p,k}^\star| < \varepsilon_\kappa, \\ \lim\limits_{r\to\infty} V_k^{(r)}, \quad V_k^{(r+1)} = \min\left\{ v_{x,\max}, \sqrt{\frac{a_N^{\max}(V_k^{(r)})}{|\kappa_{p,k}^\star|}} \right\}, \end{cases} \tag{101}$$

initialized with $V_k^{(0)} = v_{x,\max}$. For any candidate speed V at station k, define

$$a_{N,k}(V) = |\kappa_{p,k}^\star| V^2, \qquad \chi_k(V) = \left[1 - \left(\frac{a_{N,k}(V)}{a_N^{\max}(V)} \right)^p \right]_+^{1/p}, \tag{102}$$

where $[\cdot]_+ = \max\{0, \cdot\}$ and p is the GGV exponent in (19). The available forward and braking accelerations are

$$a_{T,k}^+(V) = a_T^{+,\max}(V)\chi_k(V) + a_{\rm drag}(V), \tag{103}$$

$$a_{T,k}^-(V) = a_T^{-,\max}(V)\chi_k(V) - a_{\rm drag}(V), \tag{104}$$

where $a_T^{+,\max}$ and $a_T^{-,\max}$ are the positive (driving) and negative (braking) envelopes of the symmetric limit $a_T^{\max}$ in (19), calibrated separately to account for the asymmetric powertrain and brake characteristics. Under the sign convention of (21), $a_{\rm drag}$ is usually negative at racing speeds; hence it reduces forward acceleration and assists braking.

The forward sweep starts from the measured speed,

$$V_0^f = \min\{\max(V_e, V_{\min}), V_{{\rm lim},0}\}, \tag{105}$$

and propagates

$$V_{k+1}^f = {\rm clip}_{[V_{\min},\, V_{{\rm lim},k+1}]} \left(\sqrt{(V_k^f)^2 + 2\Delta\ell_k\, a_{T,k}^+(V_k^f)} \right), \quad k = 0, \dots, N-2. \tag{106}$$

The backward sweep is initialized by

$$V_{N-1}^b = \begin{cases} \min\{V_T, V_{{\rm lim},N-1}\}, & \text{if a terminal speed is enforced,} \\ \min\{V_{N-1}^f, V_{{\rm lim},N-1}\}, & \text{otherwise,} \end{cases} \tag{107}$$

where V_T is a downstream speed constraint (e.g. from a sharp corner beyond the horizon or from $V_{{\rm cap},t}$). and propagates

$$V_k^b = {\rm clip}_{[V_{\min},\, V_{{\rm lim},k}]} \left(\sqrt{(V_{k+1}^b)^2 + 2\Delta\ell_k\, a_{T,k+1}^-(V_{k+1}^b)} \right), \quad k = N-2, \dots, 0. \tag{108}$$

The final speed surface is the pointwise feasible envelope

$$V_k^\star = \min\{V_k^f, V_k^b, V_{{\rm lim},k}, V_{{\rm cap},t}\}. \tag{109}$$

The accelerations stored with the local reference are

$$a^{\star}_{T,k} = \frac{(V^{\star}_{k+1})^2 - (V^{\star}_k)^2}{2\Delta\ell_k}, \qquad a^{\star}_{N,k} = \kappa^{\star}_{p,k}(V^{\star}_k)^2. \tag{110}$$

Finally, the time stamps are reconstructed by trapezoidal time integration,

$$\tau^r_0 = 0, \qquad \tau^r_{k+1} = \tau^r_k + \frac{2\Delta\ell_k}{V^{\star}_k + V^{\star}_{k+1} + \varepsilon_v}, \tag{111}$$

and the planner publishes

$$\mathcal{R}_t = \{(X^r_k, Y^r_k, \psi^r_k, \kappa^r_{p,k}, V^r_k, \tau^r_k)\}_{k=0}^{N_{\text{out}}-1}, \tag{112}$$

after resampling the path, speed and time arrays to the controller cadence ($N_{\text{out}} = N_c + 1$ points at intervals T_c).

5 Model Predictive Contouring Control

The control layer tracks the path-speed reference $\mathcal{R}_t$ from (112) by solving a receding-horizon nonlinear program at the actuation rate $1/T_c$ ($T_c = T_s = 0.05\,\text{s}$). A standard path-tracking MPC penalizes Cartesian deviations $(x-x^r)^2 + (y-y^r)^2$; this couples longitudinal timing to lateral accuracy and causes aggressive re-plans when the tactical layer shifts the corridor. By contrast, the MPCC formulation decomposes the error into a *contouring* component (normal to the path) and a *lag* component (tangential to the path), and penalizes only the former. Longitudinal timing is recovered through a speed-reference term, so a corridor-side switch only changes the reference sequence without requiring controller re-initialization.

At control instant t the current vehicle pose defines the local origin. The state is the planar pose $\boldsymbol{\chi}_k = [x_k, y_k, \theta_k]^\top$ (with θ_k denoting yaw relative to the local frame, corresponding to ψ in the global model of Sec. 2) and the input is the commanded speed and front-wheel steering angle $\boldsymbol{\nu}_k = [V_k, \delta_k]^\top$. The initial condition is

$$\boldsymbol{\chi}_{0|t} = \mathbf{0}, \qquad \boldsymbol{\nu}_{-1|t} = \left[V_{-1|t},\ \delta_{-1|t}\right]^\top, \tag{113}$$

where $\boldsymbol{\nu}_{-1|t}$ is the command applied in the previous cycle, retained for rate-constraint enforcement. The prediction model is a kinematic bicycle with an identified nonlinear yaw-rate map,

$$\boldsymbol{\chi}_{k+1|t} = \boldsymbol{\chi}_{k|t} + T_c \begin{bmatrix} V_{k|t}\cos\theta_{k|t} \\ V_{k|t}\sin\theta_{k|t} \\ \Omega(V_{k|t}, \delta_{k|t}) \end{bmatrix}, \tag{114}$$

$$\Omega(V,\delta) = \frac{\tan(a_\delta\,\delta + b_\delta)}{L_c}\,(c_V V + d_V), \tag{115}$$

where L_c is the wheelbase and $\{a_\delta, b_\delta, c_V, d_V\}$ are identified from closed-loop vehicle data using least-squares regression on measured yaw rate, speed, and steering angle. The model is deliberately low-dimensional: the GGV speed profiler in Sec. 4 has already ensured that the reference speed is dynamically feasible, so the controller need only reject pose disturbances at the actuation rate $1/T_c$.

The reference sequence is obtained by transforming the planner output into the local frame:

$$\bar{\boldsymbol{r}}_{k|t} = \left[x^r_k,\ y^r_k,\ \theta^r_k,\ V^r_k,\ \delta^r_k\right]^\top, \qquad k = 1,\dots,N_c, \tag{116}$$

where the reference steering is reconstructed from the planned path curvature $\kappa^r_{p,k}$ via the inverse kinematic relation

$$\delta^r_k = \arctan\!\left(\frac{L_c\,\kappa^r_{p,k}\,V^r_k}{V^r_k + \varepsilon_v}\right). \tag{117}$$

For each prediction step, define the position error in the moving tangent–normal frame of the reference path. Let $\boldsymbol{t}^r_k = [\cos\theta^r_k, \sin\theta^r_k]^\top$ and $\boldsymbol{n}^r_k = [-\sin\theta^r_k, \cos\theta^r_k]^\top$ be the unit tangent and normal at step k. The contouring error (normal deviation), lag error (tangential deviation), heading error, and input-tracking error are

$$e_{c,k|t} = (\boldsymbol{n}^r_{k|t})^\top \left([x_{k|t} - x^r_k,\ y_{k|t} - y^r_k]^\top\right), \tag{118}$$

$$e_{\ell,k|t} = (\boldsymbol{t}^r_{k|t})^\top \left([x_{k|t} - x^r_k,\ y_{k|t} - y^r_k]^\top\right), \tag{119}$$

$$e_{\psi,k|t} = \mathrm{wrap}(\theta_{k|t} - \theta^r_k), \tag{120}$$
$$\boldsymbol{e}_{\nu,k|t} = \boldsymbol{\nu}_{k|t} - \boldsymbol{\nu}^r_k. \tag{121}$$

The contouring error e_c measures the lateral deviation from the planned path and is the primary tracking objective. The lag error e_ℓ is not penalized in the cost; longitudinal timing is handled implicitly through the speed reference in $\boldsymbol{e}_\nu$. This choice avoids forcing the vehicle to chase replanned time-stamps when the tactical layer shifts the corridor.

The stage cost and the complete optimization problem are

$$\begin{aligned}
\min_{\mathcal{U}_t}\ & q_{\Delta\delta}\big(\delta_{0|t} - \delta_{-1|t}\big)^2 + \sum_{k=0}^{N_c-1}\Big[q_c\, e^2_{c,k+1|t} + q_\psi\, e^2_{\psi,k+1|t} + \boldsymbol{e}^\top_{\nu,k|t} R_\nu\, \boldsymbol{e}_{\nu,k|t}\Big] \\
\text{s.t. }\ & \boldsymbol{\chi}_{0|t} = \mathbf{0}, \\
& \boldsymbol{\chi}_{k+1|t} = f^{\mathrm{kin}}_d(\boldsymbol{\chi}_{k|t}, \boldsymbol{\nu}_{k|t}), \quad k = 0, \ldots, N_c-1, \\
& V_{\min} \le V_{k|t} \le v_{x,\max}, \\
& |\delta_{k|t}| \le \delta_{\max}, \\
& |V_{k|t} - V_{k-1|t}| \le \Delta V_{\max}, \\
& |\delta_{k|t} - \delta_{k-1|t}| \le \dot{\delta}_{\max}\, T_c,
\end{aligned} \tag{122}$$

where $R_\nu = \mathrm{diag}(r_V, r_\delta)$, and $f^{\mathrm{kin}}_d(\cdot)$ denotes the kinematic discrete dynamics in (114)–(115). The weight $q_{\Delta\delta}$ penalizes the first steering increment to enforce actuator continuity across cycles. The speed and steering rate constraints prevent saturation and ensure smooth transitions during corridor-side switches initiated by the tactical layer.

The NLP is solved in condensed form with $2N_c$ decision variables by a warm-started interior-point method. The receding-horizon policy applies only $(V^\star_{0|t}, \delta^\star_{0|t})$ and re-solves at the next cycle. The actuator mapping converts this pair into the physical command vector

$$\boldsymbol{u}_{\mathrm{act}} = \mathcal{M}(V^\star_{0|t}, \delta^\star_{0|t}) = \big[T_{\mathrm{cmd}},\ P_{b,\mathrm{cmd}},\ S_{\mathrm{cmd}},\ g_{\mathrm{cmd}}\big]^\top, \tag{123}$$

where $T_{\mathrm{cmd}} \in [0, 100]$ is the throttle percentage, $P_{b,\mathrm{cmd}} \in [0, P_{b,\max}]$ is the brake pressure, $S_{\mathrm{cmd}} \in [S_{\min}, S_{\max}]$ is the steering actuator position, and $g_{\mathrm{cmd}} \in \{1, \ldots, G\}$ is the gear. The longitudinal force request driving the throttle/brake split is

$$F^{\mathrm{req}}_x = \underbrace{c_{d,2}V^2 + c_{d,1}V + c_{d,0} + m\frac{V_{\mathrm{la}} - V^\star_{0|t}}{T_{\mathrm{la}}}}_{F_{\mathrm{ff}}} + \underbrace{K^V_p(V^\star_{0|t} - V_{\mathrm{meas}}) + K^V_i I_V}_{F_{\mathrm{fb}}}, \tag{124}$$

where $V = V_{\mathrm{meas}}$ is the current measured speed, V_{la} is a look-ahead reference speed interpolated T_{la} seconds ahead on the speed profile, I_V is the integrator state of the speed error, and K^V_p, K^V_i are PI gains. The allocation rule is

$$T_{\mathrm{cmd}} = \begin{cases} \min\big\{100\, F^{\mathrm{req}}_x / F_{\max}(\omega_e, g),\ T_{\mathrm{tc}}\big\}, & F^{\mathrm{req}}_x > 0, \\ 0, & \text{otherwise}, \end{cases} \tag{125}$$

$$P_{b,\mathrm{cmd}} = \begin{cases} 0, & F^{\mathrm{req}}_x > 0, \\ \mathrm{clip}\left(\dfrac{-F^{\mathrm{req}}_x\, r_w}{2k_b},\ 0,\ P_{b,\max}\right), & \text{otherwise}, \end{cases} \tag{126}$$

where T_{tc} is the traction-control limit and k_b converts pressure to wheel torque. The steering actuator is mapped as $S_{\mathrm{cmd}} = r_s(\delta^\star_{0|t} + \delta_{\mathrm{off}})$ with calibrated ratio r_s and offset δ_{off}.

6 Simulation Results

6.1 High-Fidelity A2RL Simulation Environment

The closed-loop experiments were carried out in the high-fidelity simulator provided by the Abu Dhabi Autonomous Racing League (A2RL) [37], an international competition for full-scale autonomous racecars, which embeds a detailed Dallara EAV24 vehicle-dynamics model together with the Yas Marina track and environment model. Internally, the simulator and the decision stack communicate over a ROS2 platform, while the final actuator commands are delivered through a virtual CAN channel that mirrors the race-vehicle interface. Sensor states, opponent states, and race-control flags are received from the simulator, and throttle, brake, steering, and gear commands are returned through the same loop. Because the decision stack runs as an external real-time process and all timing must survive the message transport layer, failures caused by delayed tactical updates, infeasible planner outputs, or controller fallback are visible at the same closed-loop interface that would be used by the race vehicle. An overview of the A2RL experimental setup is shown in Figure 4.

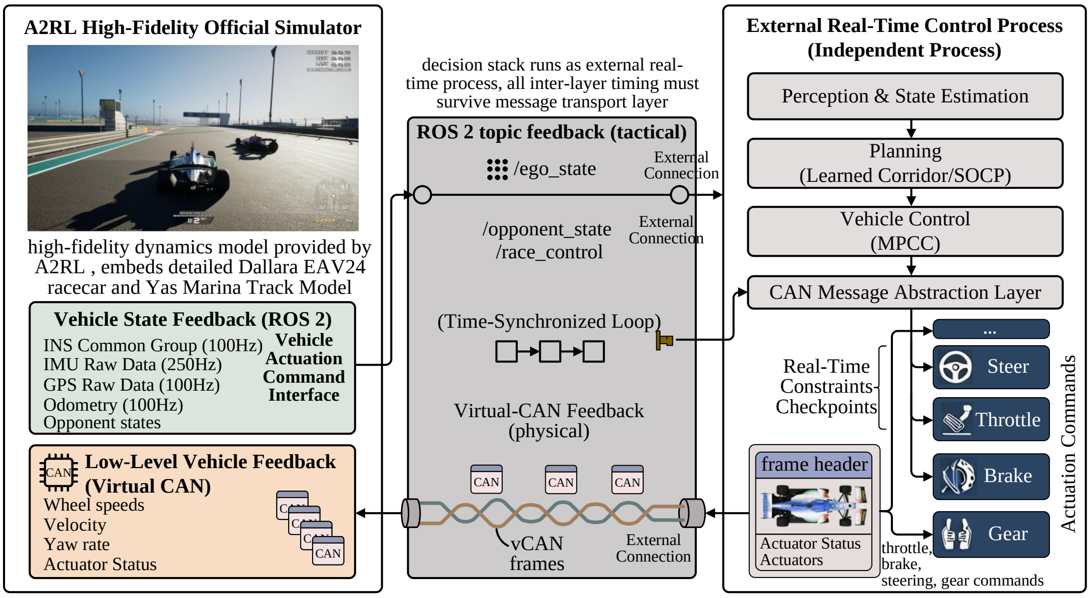


Figure 4: A2RL high-fidelity simulation environment.

6.2 *Parameter Configuration and Scenario Design*

All numerical parameters used in the experiments are summarized in Table 1. The table intentionally reports only symbols and values so that the implementation settings can be inspected without interrupting the method derivations in the preceding sections.

Three overtaking scenarios are designed to cover complementary operating regimes that a competitive racing system must handle:

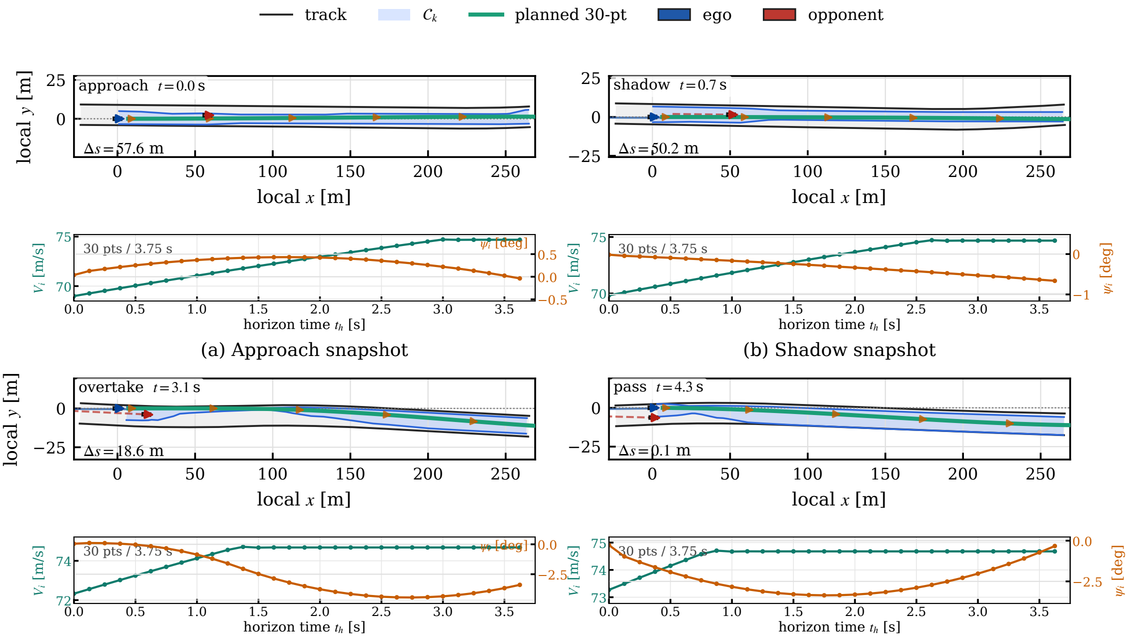
- **Scenario I (High-Speed Straight):** A pure straight section ($\kappa \approx 0$, minimum radius $> 900\,\mathrm{m}$) where both vehicles operate above $68\,\mathrm{m/s}$. This case tests whether the hierarchy can exploit a short high-speed closure window where decision latency directly determines success or failure.
- **Scenario II (Sharp-Corner):** A tight cornering section with minimum radius $\approx 10\,\mathrm{m}$ ($\kappa_{\max} = 0.098\,\mathrm{m}^{-1}$) and speed variation from 12.5 to $65.8\,\mathrm{m/s}$. This case tests whether the corridor remains feasible through severe curvature where the GGV envelope restricts available acceleration and lateral space is tightest.
- **Scenario III (Mixed Straight-Corner):** A mixed section combining straights and corners ($\kappa_{\max} = 0.037\,\mathrm{m}^{-1}$) against an equal-speed opponent. This case tests adversarial passing when the ego must generate its own advantage from track-geometry exploitation rather than an initial speed surplus.

6.3 *Scenario I: High-Speed Straight*

The ego vehicle enters the engagement with a large closing-speed advantage and must convert this advantage into a feasible lateral corridor before reaching the opponent. The run completes the pass with $\min \Delta s = -0.48\,\mathrm{m}$, $\Delta t_{\text{pass}} = 1.13\,\mathrm{s}$ after entering OVERTAKE, $\max V = 73.10\,\mathrm{m/s}$, $\mathrm{RMS}(e_c) = 0.02\,\mathrm{m}$, and $\mathrm{RMS}(e_V) = 0.26\,\mathrm{m/s}$.

Table 1: Parameter setting summary.

Sym.	Val.	Sym.	Val.	Sym.	Val.	Sym.	Val.	Sym.	Val.
m	742 kg	l_f, l_r	1.75, 1.35 m	I_z	1500 kg m^2	C_f, C_r	2.30, 2.40×10^5 N/rad	μ	1.20
$\delta_{\max}$	0.342 rad	$\dot{\delta}_{\max}$	0.5 rad/s	$[\underline{a}, \overline{a}]$	$[-5, 8]$ m/s^2	$v_{x,\max}$	90 m/s	ε_v	0.5 m/s
$c_{d,2}$	-1.04765 N s^2/m^2	$c_{d,1}$	47.5 N s/m	$c_{d,0}$	-1609.73 N	p	0.75	T_s	0.05 s
V_{char}	80 m/s	n_{char}	8 m	a_{char}	20 m/s^2	Δs_{char}	50 m	$\Delta V_{s,\text{char}}$	30 m/s
g_{char}	50 m	$\tau_{\text{tc,max}}$	30 s	ε_τ	0.01 m/s	L_t^{cap}	[1.5, 6.0]	R_t^{cap}	[1.5, 6.0]
d_{safe}	[0.2, 0.8]	V_{cap}	[15, 90]	g_{chase}	[18, 24]	g_{ot}	[16, 24]	g_{abort}	[28, 45]
b_{lat}	$[-1.5, 1.5]$	w_{safe}	[1, 5]	q_{side}	$[-1, 1]$	$g_{\text{def}}^{\text{front}}$	25 m	$g_{\text{def}}^{\text{rear}}$	20 m
Δn_{safe}	2.5 m	$g_{\text{interlock}}$	6 m	T_{imm}	30 s	w_{ot}	30	w_{ovb}	-20
w_{col}	-100	w_{lat}	-2	w_{crush}	-5	w_{app}	3	w_{sm}	-0.1
w_{gs}	-0.5	w_{sbs}	0.5	w_{wide}	0.25	α_Q	3×10^{-4}	$\lvert\mathcal{D}\rvert$	10^5
B	256	γ	0.99	τ	5×10^{-3}	f_{feat}	51→256→256	$f_{\pi,Q}$	[256, 256]
N_q	25	K	20	d_p	300 m	N_{sta}	150	Δs_{sta}	2 m
p_{fade}	3	S_{fade}	50 m	W_{fun}	2.5 m	W_{near}	5 m	λ_{side}	0.6
$L_{\min}^{\text{corr}}$	3 m	β	0.2	k_{blend}	9	d_{rmp}	40 m	k_{sr}	15
w_ω	10	$w_{n,T}$	10	$w_{\xi,T}$	100	$V_{\min}$	0.1 m/s	ε_κ	10^{-8}
$\varepsilon_n^{\text{sl}}$	0.1 m	$\varepsilon_\xi^{\text{sl}}$	0.1°	$N_{\text{iter}}^{\text{path}}$	5	q_c	50	q_ψ	5
r_V	1	r_δ	0.05	$q_{\Delta\delta}$	30	$\Delta V_{\max}$	5 m/s	w_ϵ	1000
N_c	15	T_c	0.05 s	$T_c N_c$	0.75 s				

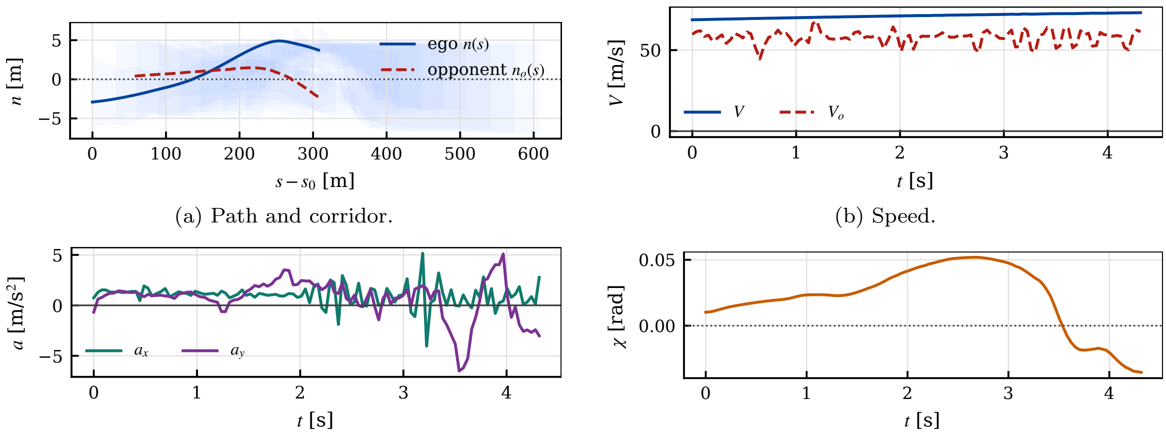


(a) Approach snapshot (b) Shadow snapshot

(c) Overtake snapshot (d) Pass snapshot

Figure 5: Scenario I engagement snapshots. The four representative instants show the planned tactical corridor and the local trajectory during approach, shadowing, commitment, and pass completion.

Figure 5 shows that the tactical decision is made before the vehicles become geometrically interlocked. In the first snapshot, the ego vehicle remains on a high-speed closure trajectory while the planner keeps the corridor centered enough to preserve recoverability. The second snapshot corresponds to the Shadow phase: the tactical layer guides the ego vehicle toward the opponent's rear-quarter on the selected pass side, biasing the corridor asymmetrically to pre-position for the upcoming overtake while retaining a conservative fallback around the raceline. In the third snapshot, the FSM has committed to Overtake; the corridor carver excludes the opponent footprint and forces the SOCP path to occupy the free side of the track. The final snapshot shows the pass completion, where the planned path has cleared the opponent without a discontinuous lateral jump. This progression is important because the learned component never commands steering or acceleration directly; it only reshapes the feasible set that the deterministic planner and MPCC must execute.

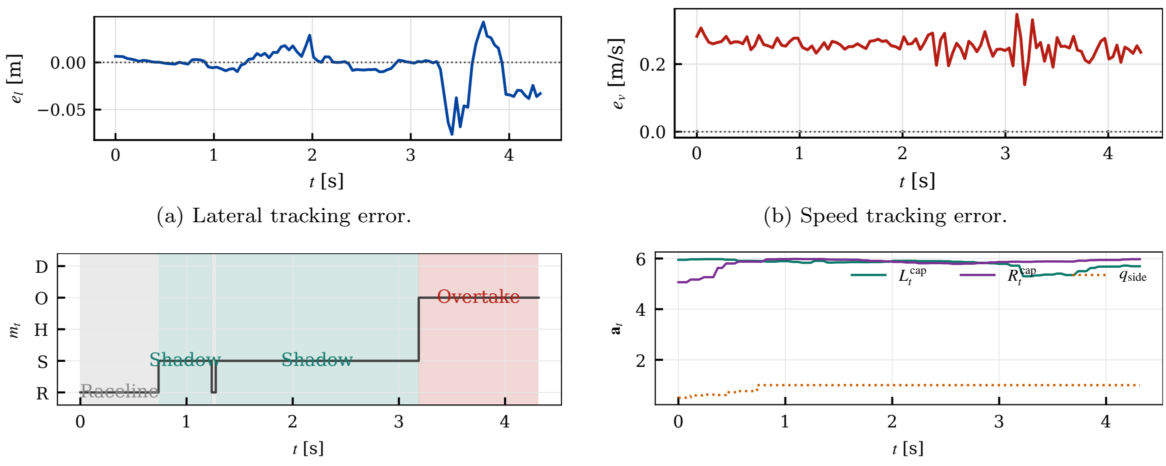


(a) Path and corridor. (b) Speed.

(c) Acceleration. (d) Heading error.

Figure 6: Scenario I closed-loop execution states.

Figure 6 quantifies the dynamic execution of the short pass. The path plot confirms that the ego trajectory remains inside the tactical corridor during the entire engagement, so the learned side decision does not create a geometric constraint violation. The speed trace reaches 73.10 m/s, indicating that the planner preserves the fast-closure advantage rather than sacrificing it through excessive caution. The longitudinal and lateral acceleration histories remain smooth during the transition into Overtake, which shows that the GGV-constrained speed profiler and MPCC friction slack prevent the maneuver from becoming a bang–bang lateral escape. The heading-error response stays bounded while the lateral path changes side, demonstrating that the corridor transition is converted into a controllable yaw demand rather than an abrupt heading correction.

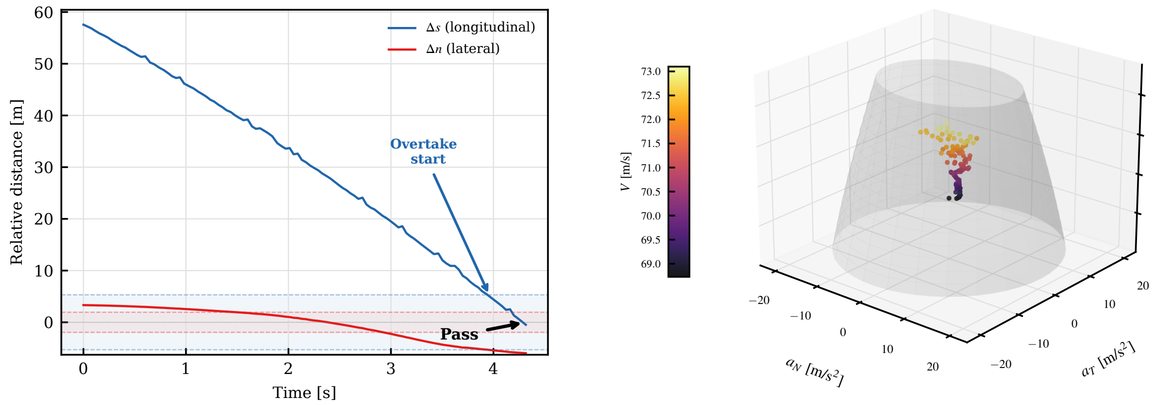


(a) Lateral tracking error. (b) Speed tracking error.

(c) FSM mode. (d) Corridor caps and side.

Figure 7: Scenario I tracking and tactical-state responses.

Figure 7 shows that the controller tracks the planner output with sub-decimeter accuracy despite the very short decision window. The lateral error has an RMS value of only 0.02 m, so the vehicle remains effectively on the planned path during the pass. The speed tracking error is also small, with $\mathrm{RMS}(e_V) = 0.26\,\mathrm{m/s}$, which confirms that the velocity cap and the GGV sweep are consistent with the MPCC's achievable acceleration envelope. The FSM mode plot shows a clean transition from Shadow—where the ego positions itself at the opponent's rear-quarter—to Overtake, while the corridor plot shows that the committed side is held during the pass rather than oscillating between alternatives. This stable side commitment is the key reason the planner can exploit the brief overtake opportunity without inducing high-frequency replanning.



(a) Collision-free verification. (b) 3-D GGV utilization.

Figure 8: Scenario I safety and performance envelope.

Figure 8(a) plots the longitudinal gap Δs and lateral gap Δn over time. Collision requires both $|\Delta s| < L_v$ and $|\Delta n| < W_v$ simultaneously; the two curves never enter their respective danger zones at the same instant, confirming collision-free execution. Figure 8(b) shows the 3-D GGV diagram where the scatter points represent actual accelerations during the engagement and the translucent surface is the speed-dependent handling envelope. On this high-speed straight, the vehicle operates well within the GGV boundary as the near-zero curvature does not demand significant lateral acceleration.

6.4 Scenario II: Sharp-Corner

The ego vehicle must remain side-by-side with the opponent through a high-curvature sequence where the GGV envelope severely limits available acceleration. The run completes the pass with min $\Delta s = -0.84\,\mathrm{m}$, $\Delta t_{\mathrm{pass}} = 8.63\,\mathrm{s}$ after entering OVERTAKE, max $V = 65.83\,\mathrm{m/s}$, RMS$(e_c) = 0.05\,\mathrm{m}$, and RMS$(e_V) = 1.82\,\mathrm{m/s}$.

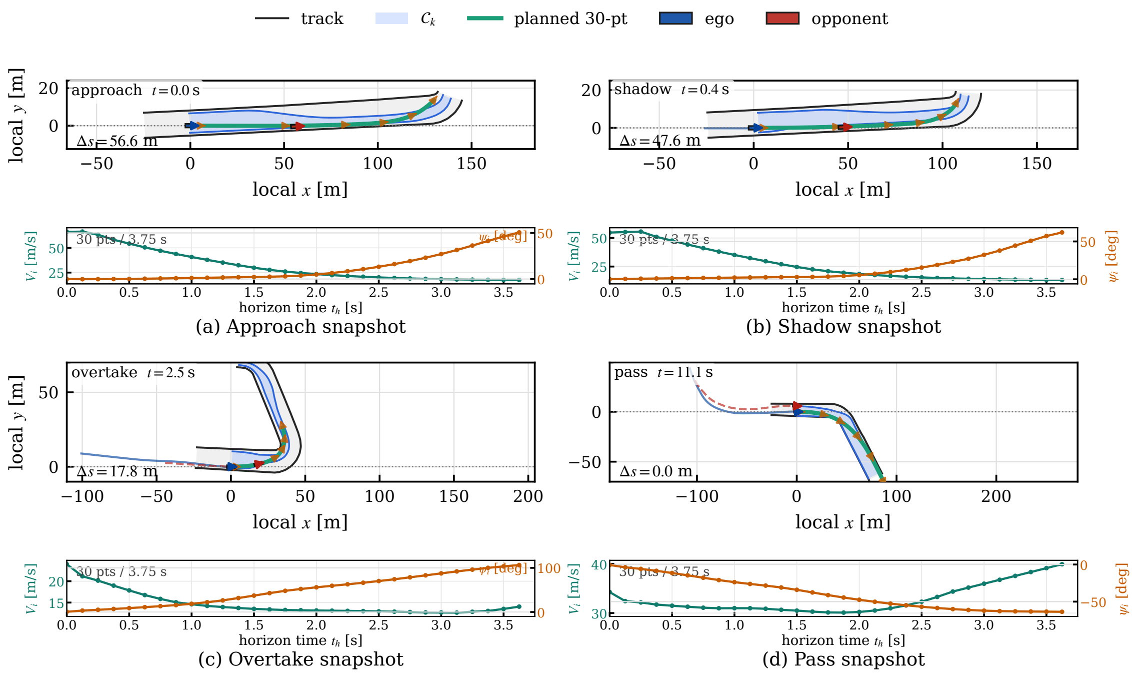
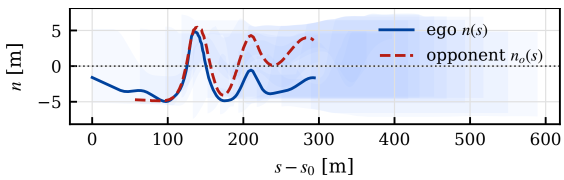
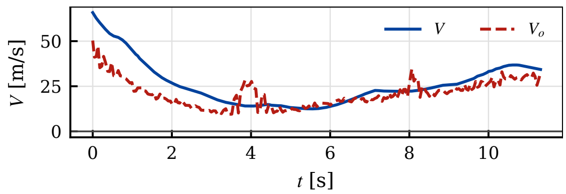


Figure 9: Scenario II engagement snapshots. The four representative instants show the planned tactical corridor and the local trajectory during the extended side-by-side overtake.

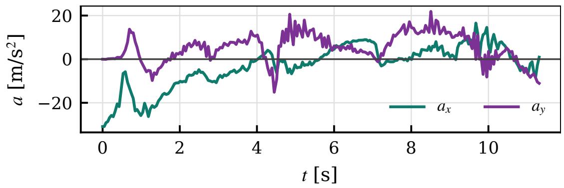
Figure 9 illustrates a qualitatively different engagement from Scenario I. The first instant shows the ego vehicle closing from behind while the corridor remains broad enough to retain both tactical options. The second instant captures the Shadow phase: the ego vehicle is guided toward the opponent's rear-quarter on the chosen pass side, with the corridor progressively narrowing to steer the planner into the optimal lateral position for the upcoming side-by-side maneuver. During the third instant, the FSM transitions to Overtake and the vehicles are side-by-side; the corridor excludes the opponent while preserving a continuous tube for the SOCP path. The fourth instant shows the ego vehicle ahead of the opponent, with the tactical layer relaxing the corridor after the pass. The sequence demonstrates that the same hierarchy handles not only short impulse-like passes, but also prolonged interactions where the corridor must remain stable for several seconds.



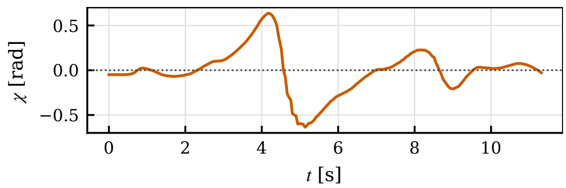
(a) Path and corridor.



(b) Speed.



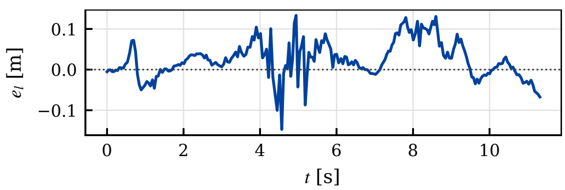
(c) Acceleration.



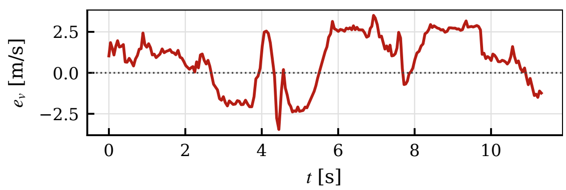
(d) Heading error.

Figure 10: Scenario II closed-loop execution states.

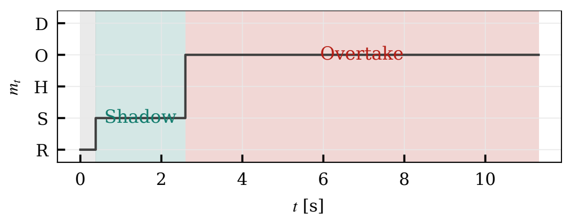
Figure 10 shows that the longer engagement is executed with a lower peak speed of 65.83 m/s, reflecting the planner's need to remain in a viable side-by-side corridor rather than complete an immediate speed-dominant pass. The path remains inside the carved corridor, which confirms that the tactical layer maintains a feasible spatial tube even when the opponent occupies the critical region for a longer time. The acceleration plot shows larger dynamic transients than Scenario I, as expected for an extended interaction, but the response remains smooth enough for the MPCC to track without friction-limit divergence. The heading response remains bounded during the side-by-side phase, indicating that the SOCP path regularization prevents the lateral maneuver from degenerating into a sharp steering correction.



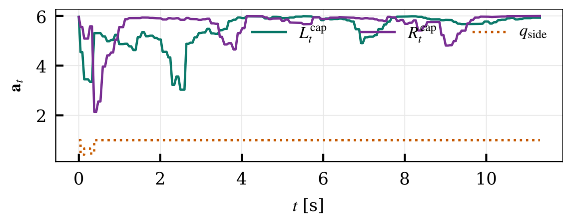
(a) Lateral tracking error.



(b) Speed tracking error.



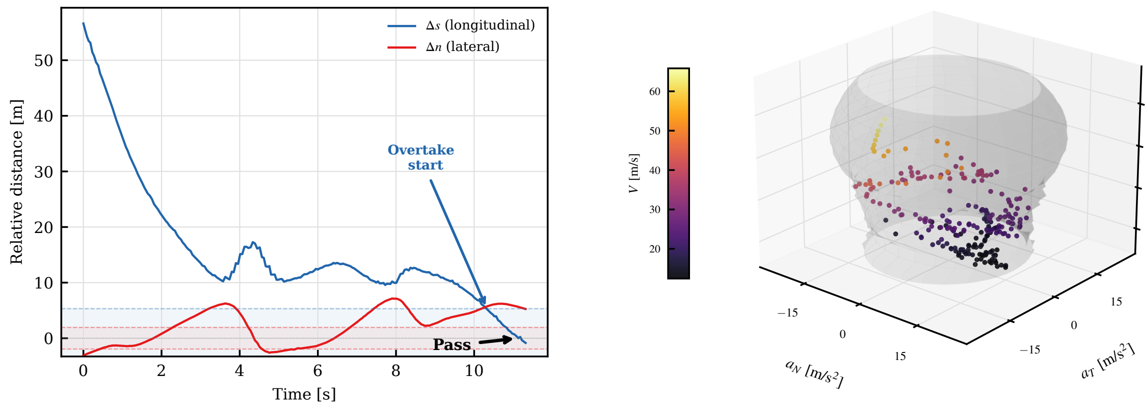
(c) FSM mode.



(d) Corridor caps and side.

Figure 11: Scenario II tracking and tactical-state responses.

Figure 11 confirms that the increased duration of the duel does not destabilize the hierarchy. The lateral RMS tracking error is 0.05 m, still well below the corridor width and therefore small relative to the available passing margin. The speed error increases to 1.82 m/s because the controller must reconcile the speed profile with a longer side-by-side geometry, but the error remains bounded and does not prevent pass completion. The FSM plot shows a persistent Overtake tenure rather than rapid mode switching, and the corridor plot shows a stable committed side throughout the decisive interval. This is the desired behavior for a learned tactical layer: once the policy selects a pass side, the downstream solvers receive a consistent feasible set long enough to execute the maneuver.



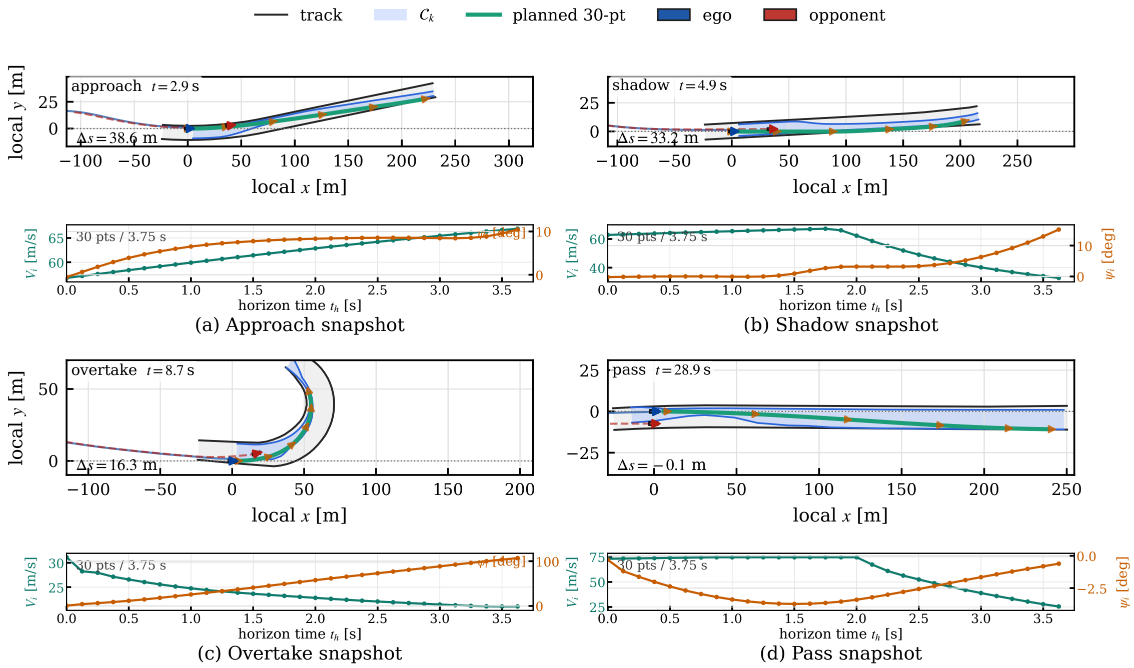
(a) Collision-free verification. (b) 3-D GGV utilization.

Figure 12: Scenario II safety and performance envelope.

Figure 12(a) confirms collision-free execution during the prolonged side-by-side phase: although Δs remains near zero for 8.6 s, the lateral gap Δn stays well outside the collision zone throughout. Figure 12(b) shows that the sharp-corner overtake pushes the vehicle close to the GGV envelope boundary, with lateral accelerations reaching the handling limit at low speeds, confirming that the planner fully exploits the available tire grip to maintain the side-by-side corridor.

6.5 Scenario III: Mixed Straight-Corner

The ego vehicle must generate its own speed advantage through corridor exploitation rather than relying on a pre-existing closure-speed gap. The analysis focuses on the decisive 29 s engagement window during which the ego vehicle identifies, commits to, and executes the overtake. The run completes the pass with $\min \Delta s = -0.12$ m, $\Delta t_{\text{eng}} = 20.13$ s from Shadow entry to pass completion, $\max V = 72.92$ m/s, $\text{RMS}(e_c) = 0.19$ m, and $\text{RMS}(e_V) = 1.62$ m/s.

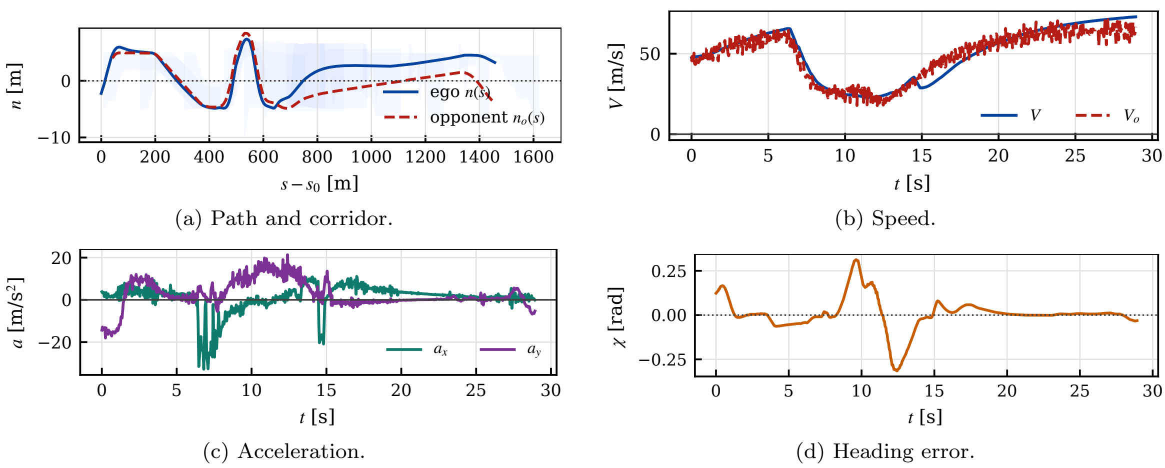


(a) Approach snapshot (b) Shadow snapshot

(c) Overtake snapshot (d) Pass snapshot

Figure 13: Scenario III engagement snapshots. The ego vehicle overtakes an equal-speed opponent through sustained corridor exploitation, requiring the FSM to progress through Shadow, Overtake, and Hold modes during the decisive engagement window.

Figure 13 reveals a fundamentally different tactical structure from the previous two scenarios. Because the opponent travels at the same nominal speed as the ego vehicle, the tactical layer cannot rely on a simple fast-closure strategy. Instead, the learned policy must identify a section of track geometry where the raceline speed surplus creates a transient advantage, commit to the corridor early enough to exploit this window, and sustain the overtake commitment through the cornering sequence. The SHADOW phase lasts 5.0 s, during which the ego vehicle is progressively steered into the opponent's rear-quarter on the committed side, building the lateral offset necessary to enter OVERTAKE with a feasible corridor. The Overtake execution then requires 8.8 s to clear the opponent and transition to Hold.



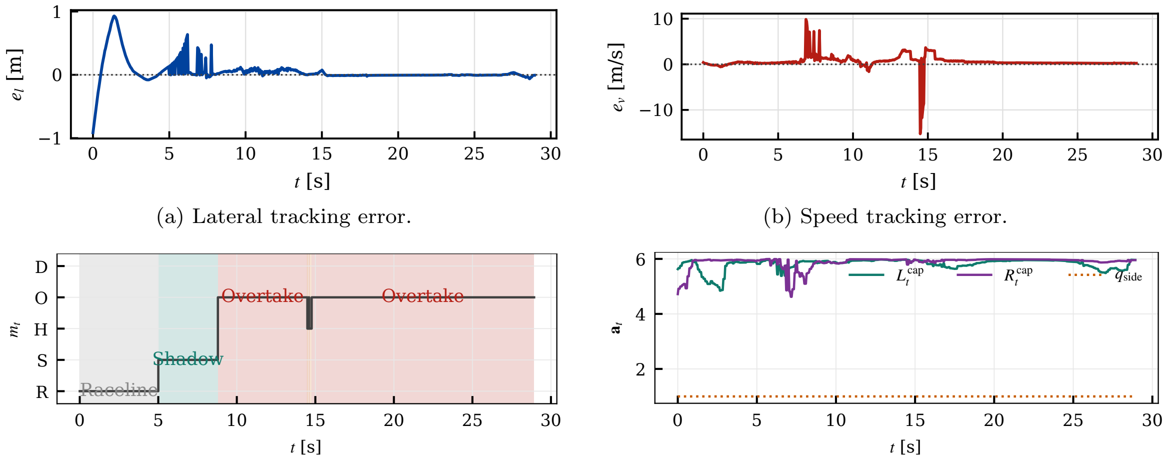
(a) Path and corridor. (b) Speed.

(c) Acceleration. (d) Heading error.

Figure 14: Scenario III closed-loop execution states.

Figure 14 shows the execution dynamics over the 29 s engagement. The path plot demonstrates that the ego trajectory commits to a lateral excursion within the corridor as the tactical layer exploits the cornering speed differential. The speed trace shows variation between 22–73 m/s, reflecting the cornering sections where the GGV profiler limits speed while the tactical layer maintains commitment to the overtake. The acceleration magnitudes reach 32.8 m/s^2

longitudinally and 21.4 m/s^2 laterally, confirming that the ego vehicle operates near the handling limits to generate the speed advantage needed against an equal-speed opponent.

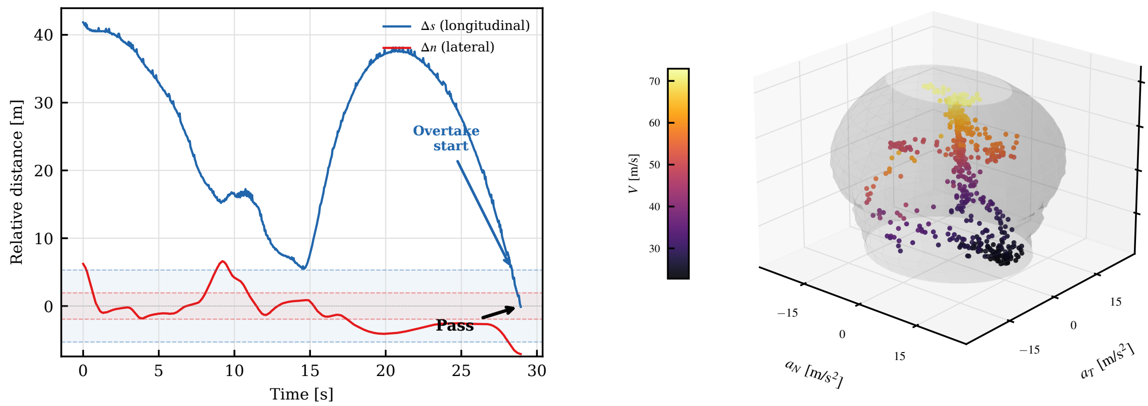


(a) Lateral tracking error. (b) Speed tracking error.

(c) FSM mode. (d) Corridor caps and side.

Figure 15: Scenario III tracking and tactical-state responses.

Figure 15 shows the tracking and FSM behavior during the adversarial engagement. The lateral RMS error is 0.19 m, larger than the two previous cases but still well within the corridor margin, confirming that the MPCC successfully tracks aggressive path references during the sustained side-by-side phase. The speed tracking error is 1.62 m/s, reflecting the controller's effort to maintain the speed advantage generated by the GGV profiler against an equal-speed adversary. The FSM mode plot shows the progression from SHADOW—where the ego is guided to the opponent's rear-quarter to establish lateral pre-positioning—through OVERTAKE to HOLD, with the mode commitment sustained throughout the decisive interval. The small longitudinal crossover margin (min $\Delta s = -0.12$ m), together with the lateral clearance shown in Figure 16, indicates that the ego vehicle generates only a limited but sufficient advantage through cornering-speed exploitation—a characteristic signature of same-speed adversarial racing where large speed surpluses are unavailable.



(a) Collision-free verification. (b) 3-D GGV utilization.

Figure 16: Scenario III safety and performance envelope.

Figure 16(a) demonstrates collision-free overtaking against the equal-speed opponent: the longitudinal gap decreases over 29 s and crosses zero only at the final pass moment, while the lateral clearance remains safely outside the collision zone throughout. Figure 16(b) reveals that this scenario demands the broadest GGV utilization—the vehicle exploits both high longitudinal acceleration on straights and significant lateral acceleration through corners, covering a wide region of the handling envelope to generate the speed advantage needed against an opponent of equal capability.

Beyond the three representative cases, the same tactical policy, planner, and controller were evaluated over 1000 closed-loop laps in the A2RL simulator. The overtake success rate is 94.7% (947/1000 laps; 95% Clopper–Pearson confidence interval [93.1%, 96.0%]), indicating that the learned tactical layer maintains consistent closed-loop performance across repeated race interactions while preserving compatibility with the model-based planner and controller. To assess sensitivity to random initialization, the TQC policy was independently trained under five different random seeds and each evaluated over the same 1000 laps; the resulting OSR across seeds is $94.3\% \pm 0.7\%$ (mean $\pm$ std over 5 seeds). This seed-level repeatability reduces the likelihood that the reported performance comes from a single favorable initialization. Equally important, the validation was performed through the virtual-CAN simulation interface, so successful laps reflect end-to-end timing consistency rather than offline trajectory feasibility alone.

Overall, the experiments support the central design principle of the paper: learning is most effective when it determines the tactical feasible set, while the planner and controller enforce geometric consistency, handling-limit speed selection, and closed-loop vehicle execution. The short-pass, long-pass, and equal-speed adversarial cases show that the architecture is not tuned to a single overtaking geometry, and the 1000-lap validation demonstrates that this separation remains reliable under repeated high-fidelity simulation.

6.6 Comparative Evaluation

To validate the proposed hierarchical game-guided corridor planning (HG-Corridor, ours), we compare it against four baselines in the same high-fidelity closed-loop A2RL simulation environment. To isolate the effect of tactical reasoning, all methods are evaluated with the same vehicle model, track model, race-control interface, SOCP path planner, speed profiler, MPCC controller, and actuator mapping. Each baseline is implemented as an alternative tactical-guidance module whose output is converted to the same corridor, side-commitment, and speed-cap interface before downstream planning; for planner-style baselines, their receding-horizon planned trajectories are projected onto this shared interface. Thus the comparison changes the tactical decision logic while keeping the closed-loop execution stack fixed.

- **Pure Raceline:** No tactical layer. The planner follows the time-optimal raceline with no opponent awareness. Overtakes occur only when the ego vehicle's speed surplus naturally creates clearance.

- **Non-Interactive MPC:** A receding-horizon tactical optimizer that treats the opponent as a dynamic obstacle whose future trajectory is predicted independently of the ego plan [38]. Avoidance constraints are imposed along the predicted opponent path, but no game-theoretic coupling or tactical mode commitment is used. This represents the predict-then-plan paradigm common in autonomous racing stacks [6, 8].

- **Rule-based GT:** A hand-tuned finite-state machine with fixed-threshold transitions ($g_{\text{chase}} = 20\,\text{m}$, $g_{\text{ot}} = 8\,\text{m}$, $g_{\text{abort}} = 35\,\text{m}$). Represents the deterministic-rules approach common in autonomous racing.

- **iLQGames:** An iterative linear-quadratic game solver [12] adapted to the racing domain. Both vehicles are modeled as rational agents in a general-sum dynamic game solved over a receding horizon via iterative best response on coupled LQ approximations. The resulting maneuver is converted to the shared tactical-guidance interface, representing the class of non-cooperative game-theoretic planners without changing the downstream execution stack.

Table 2 reports the results over 1000 closed-loop laps per method (averaged across three complementary scenarios: high-speed straight, sharp corners, and a full-lap engagement). The metrics are: overtake success rate (OSR), time-to-overtake (TTO, from engagement start to pass completion), mean per-lap minimum passing gap, collision count, average time-to-collision (Avg TTC), track-boundary violation rate, and path-feasibility ratio (PFR, fraction of cycles in which the SOCP returns a feasible solution). Defining the gap as a per-lap minimum averaged across evaluations avoids conflating the clearance statistic with the separate collision-count metric.

Figure 17 summarizes the key aggregate metrics in Table 2. HG-Corridor achieves an overtake success rate of 94.7% (95% CI [93.1%, 96.0%]), outperforming all baselines while maintaining zero observed collisions and a mean minimum passing gap of 5.83 m. Two-sided two-proportion z-tests on the OSR counts indicate that the differences between HG-Corridor and every baseline are

Table 2: Quantitative comparison against baseline algorithms over 1000 evaluation laps in the high-fidelity A2RL simulator. Best values per metric are shown in **bold**. ↑: higher is better; ↓: lower is better.

Metric	HG-Corridor (Ours)	Pure Raceline	Non-Interactive MPC	Rule-based GT	iLQGames
OSR [%] ↑	**94.7**	38.2	28.5	51.6	63.5
TTO [s] ↓	**4.82**	7.25	11.23	9.37	7.83
Mean Min Gap [m] ↑	**5.83**	1.42	3.85	2.14	2.31
Collisions ↓	**0**	21	7	48	14
Avg TTC [s] ↑	**8.52**	3.21	6.92	5.64	5.73
Track Violation [%] ↓	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
PFR ↑	**1.000**	0.988	0.998	0.987	0.995

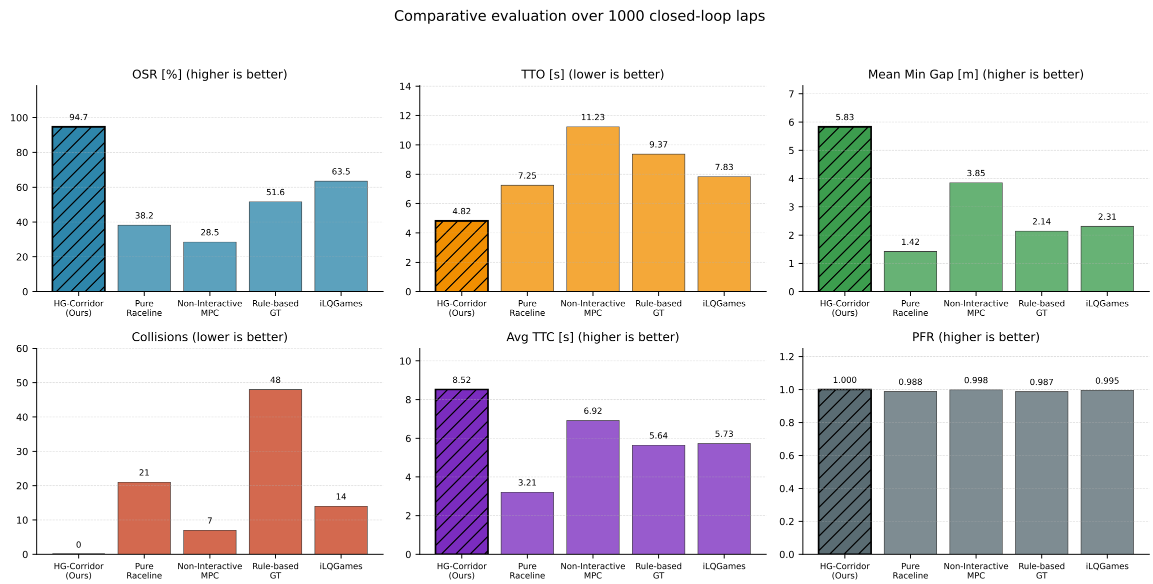


Figure 17: Aggregate comparison of key safety and performance metrics across methods using the values reported in Table 2. Hatched bars mark the best method for each metric.

statistically significant ($p < 10^{-6}$ in all cases). The strongest baseline in OSR, iLQGames, reaches 63.5% through explicit game-theoretic interaction modeling, yet still produces 14 collisions and achieves only 2.31 m mean minimum gap. This pattern indicates that explicit game solving improves overtaking relative to non-interactive planning, but does not by itself guarantee stable corridor feasibility under short decision windows or curvature-limited action spaces. HG-Corridor retains the interaction structure of the game while amortizing the tactical decision through the TQC policy, which yields side commitments that persist across planning cycles. The Non-Interactive MPC achieves the lowest OSR (28.5%) among all methods—even below the opponent-unaware Pure Raceline (38.2%). This counterintuitive result is consistent with the well-known "freezing robot" problem [40]: the predicted opponent trajectory can diverge from the realized closed-loop interaction as the ego vehicle approaches, causing avoidance constraints to become restrictive and aborting some feasible overtakes. Its low collision count (7) and large mean minimum gap (3.85 m) show that the method is conservative but competitively weak—it succeeds in keeping distance but often fails to exploit transient passing opportunities. The Rule-based GT records the highest collision count (48). This outcome is consistent with the limitation of fixed-threshold logic: a single set of gap thresholds cannot adapt to the diversity of overtaking geometries across different track sections. In HG-Corridor, the TQC policy adjusts gap thresholds, side preference, safety clearance, and corridor width from the 51-dimensional observation, which provides the adaptation missing from the hand-tuned rule set.

6.7 Ablation Study

The proposed HG-Corridor architecture integrates four key components beyond the standard SOCP+MPCC pipeline: (i) a Stackelberg–IBR game prior that provides structured interaction modeling, (ii) a hysteretic finite-state machine (FSM) that provides mode persistence, (iii) learned corridor parameters from the TQC policy, and (iv) the TQC algorithm itself (vs. standard SAC). The ablation study isolates the contribution of each component by removing it from the full system while keeping all other modules unchanged. All variants are evaluated over the same 1000 laps in the high-fidelity A2RL simulator.

- **w/o Game Prior:** The Stackelberg–IBR game module is removed. The game-prior loss and game-value auxiliary term are both disabled; the TQC policy is trained purely from the environment reward without any game-theoretic structural guidance. This tests whether the game-derived prior improves tactical timing and safety beyond what reward shaping alone provides.

- **w/o FSM:** The hysteretic FSM is bypassed. The RL policy output directly drives the corridor carver at every step without mode locking. The policy can switch between Shadow, Overtake, and Hold freely each cycle. This tests whether mode persistence is necessary for stable corridor generation.

- **w/o Learned Corridor:** The adaptive corridor parameters output by the TQC policy (including adaptive width and safety margins) are replaced with fixed nominal values (width 3.0 m, safety margin 0.5 m, and zero lateral bias). The FSM still operates but receives no learned adaptation. This tests whether the policy's continuous output contributes beyond mode selection.

- **SAC instead of TQC:** The distributional TQC policy is replaced with a standard Soft Actor-Critic trained under identical hyperparameters, network architecture, replay buffer, and reward. This isolates whether the pessimistic quantile truncation improves safety in tail-risk overtaking scenarios.

Table 3: Ablation study results. Each variant removes one component from the full HG-Corridor method. All evaluated over 1000 laps.

Metric	HG-Corridor	w/o Game Prior	w/o FSM	w/o Learned Corr.	SAC (vs. TQC)
OSR [%] ↑	**94.7**	85.4	82.3	76.1	88.5
TTO [s] ↓	**4.82**	6.38	6.71	8.94	5.53
Collisions ↓	**0**	9	12	31	5
Mean Min Gap [m] ↑	**5.83**	2.86	2.17	1.05	3.94
Avg TTC [s] ↑	**8.52**	4.93	4.31	2.86	6.17
PFR ↑	**1.000**	0.983	0.974	0.961	0.992

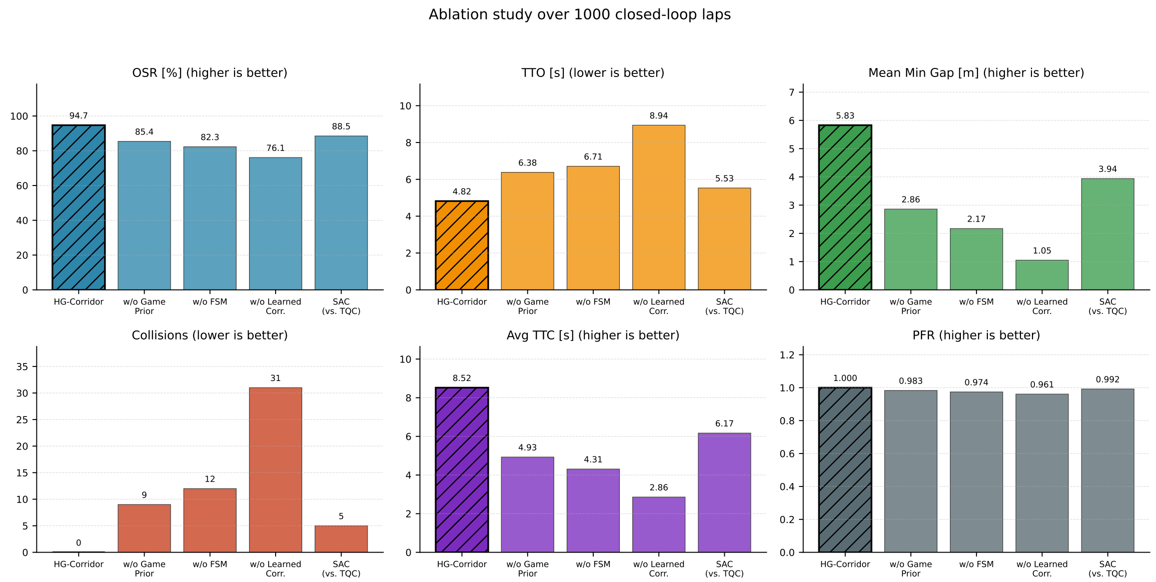


Figure 18: Aggregate ablation comparison using the values reported in Table 3. Hatched bars mark the best variant for each metric.

Table 3 and Figure 18 summarize the results. The ablation reveals a clear hierarchy of component importance (all OSR differences between the full method and each ablated variant are significant at $p < 10^{-6}$ under two-sided two-proportion z-tests):

Removing the learned corridor has the most severe impact. OSR drops from 94.7% to 76.1% (−18.6 percentage points) and collisions increase from 0 to 31. The fixed-width corridor has no mechanism to adapt to narrow track sections or opponent trajectory evolution, leading to frequent infeasible corridor states (PFR drops to 0.961) and near-misses (mean minimum gap 1.05 m, barely one vehicle width). This supports the conclusion that the policy's continuous corridor-shaping output—not just the discrete mode decision—is essential for safe high-performance overtaking.

Removing the game prior reduces OSR to 85.4% (−9.3 percentage points) and produces 9 collisions. Without the Stackelberg–IBR prior policy and game-value auxiliary loss, the TQC policy must discover overtaking timing and corridor sizing purely from environment reward. The result is later overtake commitment (TTO increases from 4.82 to 6.38 s) and more aggressive corridor sizing under uncertainty (mean minimum gap drops to 2.86 m). These results indicate that the game-theoretic structure acts as an effective inductive bias: it accelerates policy convergence toward safe, well-timed tactical decisions that the reward signal alone does not fully specify.

Removing the FSM reduces OSR to 82.3% (−12.4 percentage points) and produces 12 collisions. The mode switching rate increases 4.6× (from 0.019 to 0.087 Hz), indicating that without hysteretic locking the policy oscillates between Shadow and Overtake at mode boundaries. These side changes force abrupt corridor revisions that are hard for the SOCP planner to absorb within one planning cycle, producing transient infeasibilities and collisions near corridor reversals.

Replacing TQC with SAC achieves 88.5% OSR but introduces 5 collisions and reduces the mean minimum gap from 5.83 to 3.94 m. The SAC policy selects corridor parameters that are optimal in expectation but occasionally expose the ego vehicle to tail-risk geometries where the opponent's actual trajectory deviates from the predicted mean. TQC's pessimistic quantile truncation reduces exposure to these corridors, producing zero collisions while simultaneously improving OSR by 6.2 percentage points. In this setting, conservative tail-value estimation improves safety without sacrificing average overtaking performance.

7 Conclusion

This paper presented a hierarchical game-guided tactical corridor planning and control architecture for wheel-to-wheel autonomous racing. The framework formulates tactical interaction as a Stackelberg–IBR game and amortizes its repeated solution with a game-guided Truncated Quantile Critics policy. The learned component publishes behavior modes, side commitments, speed caps, and Frenet corridor parameters, while the deterministic FSM and corridor carver convert these outputs into auditable spatial constraints. Inside the corridor, a sparse SOCP planner generates curvature-regularized paths, a forward–backward speed sweep enforces the GGV handling envelope, and an MPCC executes the resulting path–speed reference in closed loop.

The A2RL high-fidelity simulation study shows that this division of labor is effective for short high-speed passes, prolonged cornering duels, and equal-speed adversarial engagements. Across 1000 closed-loop evaluation laps, the full architecture achieved a 94.7% overtaking success rate with no observed collisions or track-boundary violations, and the comparison and ablation studies indicate that both the game-guided policy structure and the learned corridor parameters are necessary for the reported safety and performance. The results therefore support the main design principle: learning should shape the tactical feasible set, while model-based planning and control retain responsibility for geometric consistency, handling-limit speed selection, and actuator-feasible execution. Future work will extend the framework to multi-opponent races, incorporate probabilistic opponent prediction into the corridor constraints, and validate the sim-to-track transfer on controlled Dallara EAV24 tests.

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